



北京大学

## Undergraduate Graduation Thesis

混合深度学习 GARCH 模型用于稳定币波动率  
预测

**Hybrid Deep Learning GARCH Models for  
Stablecoin Volatility Forecasting**

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# Abstract

This thesis investigates whether adding market microstructure variables to short-term stablecoin volatility forecasts improves predictive accuracy relative to a standard GARCH(1,1) benchmark. The developed hybrid model uses a feedforward neural network to adjust the GARCH parameters dynamically, conditioning on 41 microstructure inputs: centralized exchange order book data, decentralized exchange swap and liquidity metrics, and on-chain blockchain signals. It is applied to the USDC/USDT pair at five-minute frequency, using data collected from Binance, Uniswap v3, and the Ethereum blockchain over a 12-day sample in March 2026. The model's estimates are statistically insignificant, although the reason is itself informative. The benchmark GARCH(1,1) returns statistically insignificant estimates and collapses to a near-integrated process; the hybrid model's parameters converge to zero and imply a near-constant variance. Both outcomes are explained by the underlying data. The sample contains no statistically significant ARCH effects, which reflects a particularly calm trading regime over the sampling period and therefore leaves GARCH-type models unidentifiable. Despite the parameters collapsing to zero, the permutation feature importances rank the microstructure inputs in a theoretically coherent order. DEX large-trade activity, trading volume, block utilization, and liquidity concentration carry the most volatility-relevant information, an ordering that matches the price impact and arbitrage-friction mechanisms set out in the theoretical framework. The architecture thus appears to capture the relationships theory predicts. To establish forecasting value, a longer sample spanning multiple volatility regimes would be required.

**Keywords:** volatility forecasting; market microstructure; deep learning; cryptocurrency; stablecoin

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# 1. Introduction

Stablecoins have become a crucial component in the cryptocurrency ecosystem. They transform traditional assets, such as fiat currencies or commodities, into digital ones, which then serve as the primary settlement currency across centralized and decentralized exchanges, finding growing application in payments, remittances, and decentralized finance. Yet, despite being designed to uphold price stability, even the largest fiat-backed stablecoins regularly experience short-term deviations from parity. A distinguishing feature of these markets is that prices form simultaneously across centralized exchanges and decentralized automated market makers while being tied to processes on the underlying blockchain, each introducing its own set of frictions that can delay arbitrage and sustain deviations even in the absence of fundamental economic shocks.

This thesis examines whether market microstructure variables can be added to improve short-term stablecoin volatility forecasts. A hybrid model that couples a GARCH process with a neural network is conditioned on those variables and tested against the standard econometric GARCH(1,1). The hybrid architecture lets the GARCH parameters respond to the prevailing market state so that shock reactivity and volatility persistence vary with underlying conditions. The empirical application focuses on the USDC/USDT pair, using five-minute data collected across Binance, Uniswap v3, and the Ethereum blockchain over a 12-day sample in March 2026.

The thesis proceeds as follows. Chapters 2 and 3 build the institutional and theoretical foundations, covering blockchain infrastructure, stablecoin mechanics, and the microstructure of centralized and decentralized exchanges. Chapter 4 describes the data collection and variable construction. Chapter 5 introduces the GARCH benchmark. Chapter 6 develops the hybrid model and presents the empirical results and limitations. Chapter 7 concludes.

All code built for data collection, cleaning and aggregation, and training and evaluation of the GARCH benchmark and hybrid model is available on my [GitHub Repository](#).

## **2. Cryptocurrencies, Blockchain, and Stablecoins**

This chapter introduces the institutional and technical foundations underlying stablecoin trading. Because digital assets operate on blockchain networks and trade across heterogeneous markets, their economics are shaped by institutional constraints and asset-specific mechanisms. Section 2.1 describes the core features of blockchain transaction processing, including gas prices, network congestion, and asset transfers. Section 2.2 introduces the primary asset of interest, (fiat-backed) stablecoins, as pegged digital assets and discusses their defining properties. Section 2.3 elaborates on their economic use cases and market relevance.

### **2.1 Cryptocurrencies and Blockchain Infrastructure**

Unlike traditional assets, stablecoins are issued, redeemed, and settled on blockchain networks, which differ fundamentally from conventional financial infrastructure. In contrast to centralized clearing and trade execution, blockchain-based systems rely on algorithms (smart contracts), decentralized transaction validation, and fee-based transaction prioritization (Pandey et al., 2023, pp. 93, 244). These features introduce endogenous frictions into transaction execution, which affect short-term price formation and generate mechanical volatility.

To understand these mechanisms, the following subsections outline the core components of blockchain transaction processing.

#### **2.1.1 Distributed Ledgers, Blocks, and Transaction Ordering**

Blockchain technology is a peer-to-peer transaction system that employs a distributed ledger to record transactions transparently and permanently. The distributed ledger is essentially a shared database, synchronized across many geographically dispersed nodes, where each new transaction must be validated by a supermajority of nodes to ensure uniformity across the system (Pandey et al., 2023, pp. 6, 207). Each transaction recorded into a block is irreversible and publicly visible, which allows for decentralized

record-keeping of coin ownership and other digital assets. On the Ethereum blockchain, transaction ordering is economically determined by fees paid to validating nodes (validators), as higher fees create monetary incentives for validators to prioritize transactions offering greater rewards (Huberman et al., 2021; Pandey et al., 2023, p. 93).

### 2.1.2 Gas Prices, Mempool, and Congestion

Since both stablecoins of interest, USDT and USDC, are built on the Ethereum blockchain, the fee paid to validators for validating transactions is denominated in Ether and referred to as *gas* (Pandey et al., 2023, p. 93).

Events on the Ethereum blockchain are implemented through smart contracts, which are self-executing algorithms that perform a predefined operation, one of which is the addition of new transaction records to the ledger. While smart contracts greatly improve the decentralization of blockchain events, they also allow arbitrary computation on-chain, which creates two challenges for validators: computational resource consumption and the risk of unbounded execution costs. As such, users can set a maximum for how much gas is consumed, which halts the execution of a smart contract if it requires too many resources. Because these contracts consume computational resources, gas serves two more related functions: it prices resource consumption and allows users to signal priority by offering higher fees (Laurent et al., 2022). Under the EIP 1559, the total fee paid by a user is divided into a fixed base fee and a variable priority fee set by the user, which amount to the Ether offered per unit of gas and the amount of gas consumed by the operation (Buterin et al., 2019). Validators only receive the priority fee, and the base fee is burned.

$$\text{fee} = (\text{base fee} + \text{priority fee}) \times \text{gas consumed}.$$

Transactions submitted to the network but not yet confirmed are held in a waiting database known as the memory pool, or commonly abbreviated *mempool* (Pandey et al., 2023, p. 245). The degree to which transaction demand approaches network capacity is referred to as *congestion* and can be measured empirically by how full a block is. That is, the number of transactions included in a block relative to the maximum block capacity (Huberman et al., 2021). Since validators prioritize transactions offering higher fees, low-fee transactions may remain unconfirmed in the mempool for extended periods (Hossain & Altarawneh, 2025). The relationship between congestion and fees is highly nonlinear, since fees are negligible

when blocks are below roughly 50% capacity but rise sharply once congestion exceeds 80%, as users compete for faster transaction inclusion by raising their fee bids (Huberman et al., 2021).

### 2.1.3 On-Chain Events: Mints, Burns, and Transfers

On blockchain networks, changes to coin supply and ownership are implemented through smart contract events. For tokenized assets, such as ERC-20 stablecoins, the fundamental economic actions are minting and burning, which adjust supply through issuance and removal of coins, and transfers, which record changes in ownership. Those events are publicly observable and, once confirmed, irreversible.

On the Ethereum blockchain, a transfer represents the movement of tokens between addresses. In technical terms, this is implemented through a transfer function that reduces the sender's token balance and increases the recipient's, leaving total supply unchanged (Buterin, 2014; Vogelsteller & Buterin, 2015).

*Minting* refers to the creation of new tokens, which increases total supply. The minting event records a transfer from the null address `(0x0)` to the recipient while simultaneously increasing the total supply of coins tracked on the blockchain (Vogelsteller & Buterin, 2015; Zhou, 2022).

*Burning* is the inverse operation to minting. Tokens are transferred from a sender to the null address `(0x0)` and thus removed from circulation, which reduces total supply accordingly.

## 2.2 Stablecoins as Pegged Digital Assets

Stablecoins constitute a distinct class of cryptocurrencies, designed to maintain a fixed value relative to a reference asset. Reference assets range from fiat currencies to other cryptocurrencies or commodities such as gold. This thesis focuses on stablecoins pegged to the US dollar, as these dominate the market, accounting for 98.95% of total stablecoin market capitalization, and are typically backed by large pools of dollar-denominated assets, including treasury securities and bank deposits (Aldasoro et al., 2025).

Given their design objective of price stability, stablecoins present a distinct risk profile relative to other cryptocurrencies, as their value should, in principle, be insulated from broad market volatility. The following subsections discuss the economics of fiat-backed stablecoins in greater detail.

### 2.2.1 Classification of Stablecoin Designs

Stablecoins are generally classified into four categories: fiat-backed, crypto-backed, commodity-backed, and algorithmic (Mita et al., 2019). Fiat- and commodity-backed stablecoins offer the greatest price stability, but at the cost of full decentralization, as they depend on centralized issuers to hold and manage reserves. Crypto-backed coins preserve decentralization but are pegged to inherently volatile assets. Algorithmic stablecoins theoretically offer both decentralization and stability, using their own equity or a secondary token to adjust supply and maintain the peg. However, they are susceptible to death spirals and bank runs, in which a loss of confidence triggers cascading redemptions that can permanently destroy the peg and erode the coin's value (Catalini et al., 2022). A notable example of such a death spiral is the Terra-Luna collapse in 2022 (Liu et al., 2023).

### 2.2.2 Fiat-Backed Stablecoins USDT and USDC

Non-algorithmic stablecoins maintain their peg by holding reserves of the reference asset, thereby guaranteeing convertibility at the specified rate. In the case of fiat-backed stablecoins, issuers hold highly liquid bank deposits, government bonds, repurchase agreements, and high-grade commercial paper, closely resembling the reserve structure of a money market fund (Aldasoro et al., 2024). Reflecting their scale, stablecoin issuers purchased \$40B in US treasury bills in 2024, enough to even have a noticeable impact on market yields (Ahmed & Aldasoro, 2025; Aldasoro et al., 2025).

The two largest stablecoins by market capitalization are Tether (USDT), with a market cap of \$184.01B, and USD Coin (USDC), with \$78.85B, as of March 13th, 2026 (CoinMarketCap, n.d.). Both are USD-denominated fiat-backed stablecoins and together account for 90% of all USD-denominated stablecoins in May 2025 (Aldasoro et al., 2025).

Despite being the least volatile class of cryptocurrency, fiat-backed stablecoins regularly experience price fluctuations on the order of basis points (Lyons & Viswanath-Natraj, 2023). Occasionally, episodes of severe economic stress break the \$1 peg by a larger margin. Notable examples include USDC frequently trading near \$0.995 during Q3 and Q4 of 2021 and Tether reaching a low of approximately \$0.95 during Q2 2022 (Aldasoro et al., 2025).

### 2.2.3 Peg Structure, Redemption Mechanism, and Mechanical Volatility

The relative stability of fiat-backed stablecoins is not purely mechanical, but the outcome of an issuance and redemption structure designed to enforce parity through arbitrage. Taking Tether as an example: on the primary market, USDT is exchangeable for USD at a fixed 1:1 rate, subject to a deposit fee of 0.1%, a redemption fee equal to the greater of \$1,000 or 0.1% of the transaction amount, and a minimum transaction size of \$100,000 (Tether Operations Limited, n.d.-a). In this sense, the Tether treasury functions similarly to a central bank, holding US dollar-denominated assets to defend parity.

Secondary market prices, however, frequently deviate from parity and therefore require a stabilization mechanism. Lyons and Viswanath-Natraj (2023) examines decentralized arbitrage as the main stabilization mechanism. When the secondary market price of Tether falls below \$1, arbitrageurs buy Tether on exchanges and redeem it at the treasury for USD at the 1:1 rate, profiting from the spread. In contrast, when the price rises above \$1, arbitrageurs mint new Tether from the treasury and sell it on the secondary market, pushing the price back toward parity.

In the very short run, however, prices continue to deviate from parity due to several layers of execution friction. First, the deposit and redemption mechanisms carry transaction fees and minimum transaction sizes that exclude retail investors and smaller institutions from primary market arbitrage. Second, processing delays and mempool congestion introduce latency into the arbitrage process. Especially when congestion is high and participants bid up gas prices to accelerate confirmation, the cost of executing an arbitrage trade can rise sharply and can render it unprofitable, causing prices to remain deviated for longer. Third, stablecoins are simultaneously traded across heterogeneous secondary markets, both centralized and decentralized exchanges. Each venue has its own microstructural peculiarities, which introduce different frictions and can thus delay cross-market arbitrage.

As a result, short-term deviations from parity are not purely informational in origin but are partly rooted in execution frictions inherent to the issuance, redemption, and transaction settlement process. This mechanical volatility persists even in the absence of fundamental economic shocks.

## 2.3 Economic Relevance and Use Cases of Stablecoins

As a low-volatility digital asset, stablecoins have become integral to cryptocurrency trading and the broader decentralized finance ecosystem, while also finding growing relevance in traditional monetary applications as a payment system alternative that offers faster, more transparent, and lower-cost movement of funds (Catalini et al., 2022). The following subsections outline the most prominent stablecoin use cases.

### 2.3.1 Price-stable Settlement Assets in Cryptocurrency Trading

Stablecoins have become a core component of the cryptocurrency ecosystem, functioning primarily as a price-stable vehicle currency to settle transactions. By providing a stable store of value within an otherwise volatile asset class, stablecoins serve as the de facto digital equivalent of (fiat) currency. Centralized exchanges such as Binance also offer fiat-to-crypto conversion only for the largest cryptocurrencies, while decentralized exchanges such as Uniswap are crypto-only by design, requiring stablecoins to act as a stable transaction currency. Stablecoins, therefore, enable trading, settlement, and value storage within these ecosystems without requiring conversion back to traditional currency. This function extends to commodity-backed stablecoins, such as PAX Gold, which allow physical commodities to be represented as digital assets.

### 2.3.2 Payment Systems, Remittances, and Programmability

Stablecoins also have significant applications in traditional finance. Existing payment systems are slow and costly (Catalini et al., 2022), and stablecoins offer a potential alternative by enabling same-day domestic transfers, business card transactions, online settlements, and multi-recipient payments at lower cost and higher speed than current systems. Moreover, so-called institutional stablecoins, which are privately issued coins used by corporations, allow companies to transfer funds across borders and help subsidiaries manage internal liquidity more efficiently (Liao & Caramichael, 2022).

Stablecoins also find widespread use in cross-border remittances, a market characterized by high fees and commissions averaging 6.49% globally in Q1 2025, with transfers to Sub-Saharan Africa being the most expensive at 8.87% (World Bank Group, 2025). Stablecoins have the potential to substantially reduce these costs and are already widely used for cross-border transfers (Catalini et al., 2022), with combined cross-border

flows for Tether and USDC reaching between \$250B and \$400B per quarter, with volumes first exceeding \$200B in Q2 2021 (Aldasoro et al., 2025; Auer et al., 2025).

Beyond simple transfers, stablecoins support programmable payments: financial agreements tied to the fulfillment of predetermined conditions, which can unlock funds automatically upon completion of a specified action. This is particularly popular in emerging markets, where conditional cash transfer programs have a well-documented track record (Catalini et al., 2022; Parker & Todd, 2017).

### 2.3.3 The Role of Stablecoins in Decentralized Finance

Currently, the most promising application of stablecoins is in decentralized finance (DeFi). DeFi refers to the use of smart contracts in financial services, including lending, borrowing, and asset management, without relying on traditional intermediaries. DeFi protocols are also more transparent and accessible than their traditional counterparts, while also enabling faster execution (Catalini et al., 2022). Within DeFi, stablecoins serve as the primary, non-volatile asset used to store liquidity or as collateral in lending protocols. One example discussed by Fowler et al. (2020) is the Oasis smart contract, through which users can deposit funds into the algorithmic stablecoin DAI and earn the DAI Savings Rate (DSR) on their deposits. The DSR offers a predictable and stable real rate of return, which is particularly relevant during low-interest-rate environments in which the real return after inflation on traditional bank deposits may even be negative. DeFi applications of this kind have been proposed as tools for financial inclusion in developing economies, with the potential to lower holding costs, allow unbanked individuals to earn interest on savings, and connect participants in the gig economy who lack access to formal banking infrastructure or a government ID.

The use cases of stablecoins thus strongly rely on peg stability. In DeFi in particular, where smart contracts execute automatically and without human intervention, a stablecoin that loses parity can break the protocols, which are built on the assumption that the coin tracks its reference asset (Catalini et al., 2022).

Understanding the drivers of peg deviations is therefore essential both for evaluating stablecoins as reliable settlement assets and for assessing the mechanisms that induce volatility.

## 3. Market Microstructure of Stablecoin Trading

This chapter examines the market microstructure of stablecoin trading. Because stablecoins trade both on classical centralized exchanges and on novel decentralized exchanges, their respective microstructures affect price formation in different ways. Section 3.1 covers centralized exchanges and price formation through limit order books. Section 3.2 introduces decentralized exchanges based on automated market makers, describing the operation of liquidity pools and mechanisms of price determination. Section 3.3 considers the interactions between centralized exchanges, decentralized automated market makers, and on-chain events that influence arbitrage dynamics and short-term price behavior.

### 3.1 Centralized Exchanges (CEX)

Centralized exchanges, such as Binance or Coinbase, are the cryptocurrency equivalent of traditional exchanges used to trade stocks, bonds, foreign exchange, and derivatives. Similar to their traditional counterparts, centralized exchanges offer spot trading for cryptocurrencies and commonly provide additional services such as lending, borrowing, and derivative trading (Nummelin, 2022). A defining feature of centralized exchanges is that trades are not settled directly on the blockchain but are instead recorded on the exchange’s internal books (Yadav, 2023). Most centralized exchanges employ limit order books as their primary matching mechanism, though some also offer market-making services (Asgari, 2023) to bridge liquidity gaps. This is helpful particularly during off-peak hours, given that centralized exchanges operate continuously (Yadav, 2023).

The following subsections explain how prices form on centralized exchanges and how such microstructure influences prices and volatility.

#### 3.1.1 Limit Order Books and Price Formation

A *limit order book* is a centralized record of market participants’ buy and sell orders, which is used to match buyers with sellers based on the lowest available sell price and the highest available buy price. Formally, following Gould et al. (2013), an *order*  $x = (p_x, \omega_x, t_x)$

submitted at time  $t_x$  specifies a price  $p_x$  and a quantity  $\omega_x$  (negative for sell orders), and the limit order book  $\mathcal{L}(t)$  contains all such active orders  $(x(t)_n)_{n \leq |\mathcal{L}(t)|}$ . The set of active buy orders is denoted as  $\mathcal{B}(t)$ , with *best bid* price  $b(t) := \max_{x \in \mathcal{B}(t)} p_x$ , and the set of active sell orders as  $\mathcal{A}(t)$ , with *best ask* price  $a(t) := \min_{x \in \mathcal{A}(t)} p_x$ . The *bid-ask spread* is then  $s(t) = a(t) - b(t)$ . *Bid (ask) depth* at a given price  $p$ , which describes the total quantity available on the buy (sell) side at that price, can be calculated as  $n^b(p, t) := \sum_{x \in \mathcal{B}(t), p_x = p} |\omega_x|$  (respectively  $n^a(p, t) := \sum_{x \in \mathcal{A}(t), p_x = p} \omega_x$ ).

Price formation in limit order book markets is closely tied to depth. When a new order arrives, it is either immediately matched against an existing opposing order and would be called a *market order*, or it remains unmatched and enters the book as an active *limit order* (Gould et al., 2013). Taking a buy order  $x$ , there are three possible cases:

- If  $p_x \leq b(t)$ , the order is a limit order placed at or below the current best bid and does not change  $b(t)$ .
- If  $b(t) < p_x < a(t)$ , the order is a limit order placed inside the spread, which causes  $b(t)$  to increase, narrowing the spread.
- If  $p_x \geq a(t)$ , the order is a market order, since at least one opposing sell order  $y$  exists with  $p_y \leq p_x$ . The transaction executes at the lower ask price  $p_y$ . If the buy order size exceeds the available quantity at that level,  $|\omega_x| > |\omega_y|$ , subsequent sell orders at higher prices keep being matched until the remaining quantity  $|\omega_x|$  is exhausted and the new ask price is higher than the new bid price,  $a(t + \Delta t) > b(t + \Delta t)$ .

The updated bid and ask prices after execution depend on available depth:

$$b(t + \Delta t) = \max(p_y, q), \quad q = \arg \max_{k'} \left( \sum_{k=k'}^{b(t)} |n^b(k, t)| > \omega_y \right) \quad \text{after a sell market order } y,$$

$$a(t + \Delta t) = \min(p_x, q), \quad q = \arg \min_{k'} \left( \sum_{k=k'}^{a(t)} n^a(k, t) > |\omega_x| \right) \quad \text{after a buy market order } x.$$

The price displayed on the exchange is the last traded price.

This mechanism implies that price changes are a direct function of order size and the distribution of available liquidity. Fluctuations in depth and order flow can therefore generate mechanical price changes even in the absence of new fundamental information.

### 3.1.2 Liquidity, Depth, and Order Imbalance

While the mechanics of price formation in limit order books are well-defined, the magnitude of price changes critically depends on underlying market liquidity. Liquidity is commonly characterized through three dimensions: the cost of reversing a position within a short time horizon, the speed with which prices recover following a random shock, and the order size required to move prices (Kyle, 1985). The third dimension is directly tied to the depth functions  $n^b(p, t)$  and  $n^a(p, t)$ , which measure the volume available for absorption at each price level  $p$ . Empirically, market depth has been shown to be negatively correlated with subsequent short-term volatility (Ahn et al., 2001; Kalev & Duong, 2007). This relationship follows directly from the depth equations established in Section 3.1.1: shallow order books amplify price changes because incoming orders consume liquidity across multiple price levels before being fully matched (Gould et al., 2013).

Order imbalance, which describes the difference between the quantity of all buy and sell orders in the book (Cao et al., 2009), has similarly been shown to predict short-term price movements (Boehmer & Wu, 2008; Cao et al., 2009; Chordia & Subrahmanyam, 2004; Chordia et al., 2002), as traders will eventually convert their limit orders into market orders. Cao et al. (2009) further demonstrate that at sub-five-minute intervals, large order book imbalances significantly amplify short-term price movements.

## 3.2 Decentralized Exchanges (DEX)

In contrast to centralized exchanges, decentralized exchanges rely on automated market maker (AMM) protocols, which determine prices through a deterministic function rather than order matching. Limit order books are replaced by liquidity pools, in which liquidity providers deposit token pairs in exchange for a proportional share of the trading fees generated by the pool (Binance, n.d.). Unlike centralized exchanges, decentralized exchanges operate directly on the blockchain (Barbon & Ranaldo, 2024).

This combination of deterministic pricing, liquidity pools, and on-chain settlement produces a microstructure that differs from centralized exchanges. The following subsections explain how prices form on decentralized exchanges and how their microstructure affects price dynamics and volatility.

### 3.2.1 Automated Market Makers and Constant Function Pricing

*Automated market makers* (AMMs) are smart contracts that operate on pools composed of two tokens. Traders interact directly with the pool rather than submitting orders to a book, and prices are determined by the relative quantities of assets held in the pool (Uniswap Labs, n.d.-d). Constant function market makers (CFMMs), such as Uniswap, which is the protocol examined in this thesis, determine prices according to a simple function. For a pool containing  $x$  units of token 0 and  $y$  units of token 1, the AMM enforces the constant product rule  $x \times y = k$ , where  $k$  is a constant that must be preserved after every transaction (Jeiwan, n.d.).

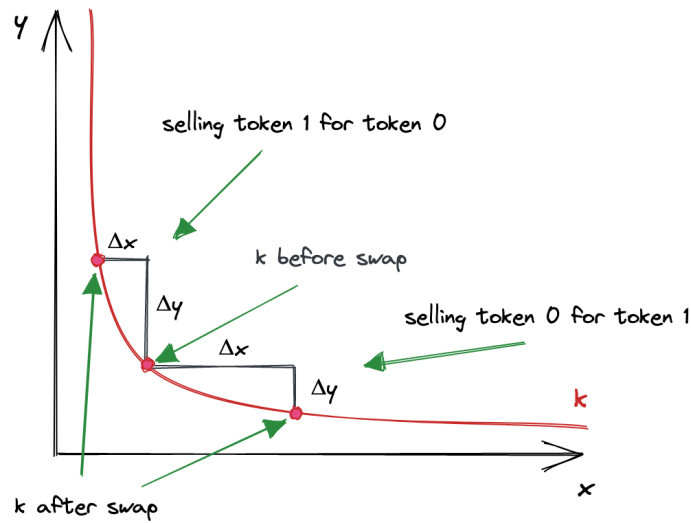
When a trader exchanges a quantity  $\Delta x$  of token 0 for a quantity  $\Delta y$  of token 1, a fee  $r$  is levied on the input amount, and the AMM equation becomes  $(x + r\Delta x)(y - \Delta y) = k$ . The spot prices  $P_x = y/x$  and  $P_y = x/y$  imply the following output and input quantities:

$$\Delta y = \frac{y r \Delta x}{x + r \Delta x},$$

$$\Delta x = \frac{x \Delta y}{r(y - \Delta y)}.$$

This mechanic is illustrated by Figure 3.1.

Figure 3.1: Constant product curve illustrating price impact for a two-token AMM pool.  
Source: (Jeiwan, n.d.)



These relationships show that price changes are determined by trade size relative to pool depth: larger trades consume a greater share of the pool's reserves, leading to a more

pronounced price impact. As a result, volatility arises endogenously from trading activity, even in the absence of new fundamental information.

### 3.2.2 Liquidity Provision in Automated Market Makers

Liquidity in AMMs is supplied externally by liquidity providers (LPs), who deposit token pairs into the pool and in return receive a share of the trading fees proportional to their contribution (Zhu et al., 2025). By committing tokens to the pool, LPs are exposed to price and demand changes associated with the pool. If the price of token 0 falls, traders will exchange token 0 for token 1, causing LPs to accumulate an increasing share of the depreciating asset. If the price decline continues, LPs are progressively forced into a larger position in the falling asset and a smaller position in the appreciating one. This dynamic has the same payoff as being short both a call and a put on the underlying asset, as the LPs increase exposure to the depreciating asset while giving up exposure to the appreciating token (Lipton et al., 2024). This effect is known as impermanent loss.

As such, LPs face a trade-off between earning trading fees and bearing the risk of impermanent loss. To allow LPs to better manage this trade-off, protocols beginning with Uniswap v3 introduced multiple fee tiers (0.01%, 0.05%, 0.30%, and 1%) allowing LPs to select the compensation level appropriate for the risk of their chosen pool (Uniswap Labs, n.d.-c). Consistent with traditional market-making rationale, highly liquid and stable pairs such as stablecoin pools gravitate towards the lowest fee tier of 0.01%, while pools with lower volume or more volatile assets require higher fees to compensate LPs adequately (Uniswap Labs, n.d.-c).

Changes in liquidity provision directly affect pool depth, which in turn determines price sensitivity. Price impact and liquidity are inversely related: reductions in liquidity amplify the price impact of any given trade, while increases in liquidity dampen it.

### 3.2.3 Concentrated Liquidity and Tick-Based Pricing

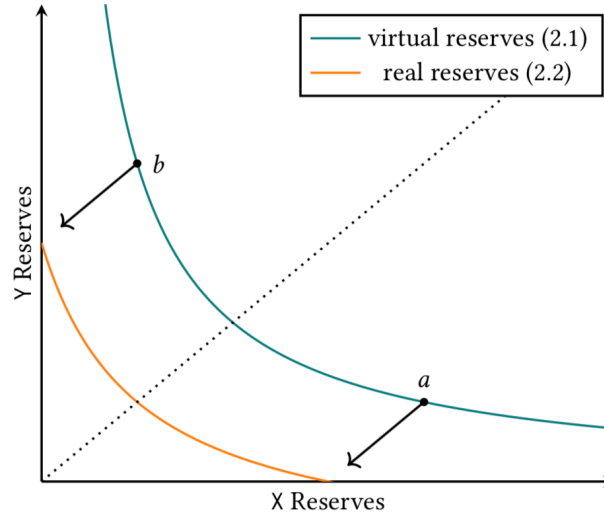
Early CFMMs, such as Uniswap v1 and v2, distribute liquidity uniformly across all possible price ranges along the entire constant product curve from zero to infinity. For trading pairs that remain close to a fixed price, such as stablecoins maintaining parity against their reference asset, this means that the vast majority of liquidity is allocated to price ranges where trading never occurs, reducing capital efficiency and lowering fee income for LPs (Uniswap Labs, n.d.-a). Uniswap v3 addresses this by introducing *concentrated liquidity*,

allowing LPs to restrict their capital to a specified price range  $[p_a, p_b]$  (Adams et al., 2021). Within this range, the position behaves as if the pool held larger reserves, which are referred to as 'virtual reserves,' as shown in Figure 3.2, making the effective curve steeper within the range and providing no liquidity outside it. The modified AMM equation for a position is

$$(x + \frac{L}{\sqrt{p_b}})(y + L\sqrt{p_a}) = L^2.$$

where  $L = \sqrt{k}$  is the liquidity parameter, and  $\sqrt{p_a}$ ,  $\sqrt{p_b}$  denote the square roots of the lower and upper price bounds of the position (Adams et al., 2021; Jaiwan, n.d.).

Figure 3.2: Virtual versus real reserves under concentrated liquidity in Uniswap v3. Source: (Adams et al., 2021)



For stablecoin pairs, LPs typically concentrate their liquidity within the range  $[0.99, 1.01]$ , as nearly all trading volume occurs near parity (Uniswap Labs, n.d.-a). This segmented liquidity structure introduces non-linear price dynamics: within an active position's range, depth is high and price impact is low, but once the price crosses a range boundary, active liquidity drops sharply, as positions concentrated within  $[0.99, 1.01]$  no longer contribute to available liquidity. Consequently, volatility may spike when prices approach or cross these boundaries, as reduced depth increases price impact per unit of trade volume.

### 3.3 Interaction Between CEX, DEX, and On-Chain Processes

Because stablecoins are simultaneously traded on centralized exchanges and decentralized protocols and settled on the blockchain, the heterogeneous microstructures across these venues give rise to differences in price formation and liquidity provision. Although the markets are connected through arbitrage, differences in execution mechanisms and transaction costs can sustain price discrepancies in the short run.

This section examines the mechanisms through which prices adjust across centralized and decentralized exchanges and the frictions that impede the speed and effectiveness of adjustment.

#### 3.3.1 Arbitrage and Cross-Market Transmission

In integrated financial markets, the law of one price dictates that the same asset should trade at the same price across all venues. This is enforced through arbitrage, which means that whenever prices diverge, traders exploit the discrepancy until it is eliminated.

Consider a market for tokens 0 and 1 with a single liquidity pool on a DEX and a single trading pair on a CEX. If the price of token 0 is higher on the CEX, an arbitrageur would sell token 0 on the CEX and use the proceeds to buy token 0 on the DEX, earning the spread as profit. The increased demand for token 0 on the DEX shifts the pool composition towards holding more of token 1 and less of token 0, pushing the DEX price of token 0 ( $P_0 = \text{token 1} / \text{token 0}$ , see Section 3.2.1) upward toward the CEX price.

On the CEX side, the arbitrageur issues market sell orders for token 0, which match against the best available bid orders. Each executed sell progressively consumes bid-side depth from  $b(t)$  downward, causing the best bid  $b(t + \Delta t)$  and consequently the displayed price to decline, in line with the mechanics described in Section 3.1.

Through the described arbitrage mechanisms, especially in highly traded pairs, price information is transmitted across different venues, which theoretically should equalize prices after deviations. However, those adjustments are not instantaneous. Temporary deviations can persist due to frictions, which gives rise to short-term volatility.

#### 3.3.2 Gas Costs and Execution Frictions

Arbitrage between centralized and decentralized exchanges is subject to execution frictions, most notably transaction costs, which consist of exchange fees and gas, and network

congestion. Importantly, arbitrage is only profitable when the spread between the CEX and DEX prices exceeds the total cost of execution, as below this threshold, arbitrage becomes unprofitable despite price differences being observable.

As described in Section 2.1.2, gas is charged when executing a smart contract event such as a DEX trade and consists of a user-set gas price multiplied by the computational resources consumed. When a large spread between venues emerges, multiple arbitrageurs simultaneously attempt to capture it, submitting competing transactions to the mempool. Since only the first handful of arbitrageurs to be validated capture the profit, competing traders engage in priority gas auctions, bidding up their gas prices to be the first ones to be validated (Huberman et al., 2021). This gas price competition progressively tightens the real, post-fee spread, and pushes returns toward a no-arbitrage band. In this way, execution costs function as an endogenous competitive mechanism that limits arbitrage profitability and perversely inhibits the stabilizing role of arbitrage across markets.

As a result, short-term deviations from parity do not correct immediately. Under high network congestion, deviations may persist or even widen before arbitrage becomes viable again, which contributes to the mechanical volatility examined in this thesis.

## 4. Data and Variable Construction

This chapter describes the data collection process and the construction of the model’s inputs. Section 4.1 gives rationale for the selection of trading venues, blockchain data source, and asset pair. Section 4.2 explains the data collection procedure for CEX, DEX, and on-chain inputs. Section 4.3 presents the process and results of cleaning data.

### 4.1 Market Selection and Sample Design

The cryptocurrency ecosystem comprises numerous centralized and decentralized exchanges, all of which are distinct in size, liquidity, asset focus, and microstructure. According to a recent industry report, centralized exchanges account for approximately 86% of spot transaction volume, with decentralized exchanges processing the remaining 14% (Zhong et al., 2026). Similarly, as of the 23rd of March 2026, nine fiat-backed stablecoins exceed a market capitalization of \$1B and the next 24 are capitalized above \$100M (DefiLlama, n.d.). Given this breadth, the selection of appropriate venues for data sourcing and the stablecoin pair is important for the validity and interpretability of the empirical analysis.

This section documents the selection of markets and asset pair, which are used throughout the thesis.

#### 4.1.1 Exchange and Blockchain Selection

**Centralized exchange.** Among centralized exchanges, Binance is the dominant venue for spot cryptocurrency trading, holding a market share of 39.6% over the past six months, whereas the next largest CEX processes less than a fourth (8.2%) of that. (Zhong et al., 2026). Aside from its market position, Binance publishes historical candlestick (kline) data for spot pairs at all standard intervals, including five minutes, making it suitable for high-frequency empirical work and simplifying data collection. Binance is therefore the natural choice for the centralized exchange data source for this thesis.

**Decentralized exchange.** For the decentralized exchange, the primary candidates are ‘Uniswap’, operating on the Ethereum blockchain, and ‘PancakeSwap’, operating on the

BNB Chain, both of which have reported comparable spot trading volumes of approximately \$0.54T over the past six months. This thesis uses Uniswap for four reasons. *First*, Uniswap is the most widely studied DEX in academic literature (Barbon & Rinaldo, 2024; Lipton et al., 2024), providing a well-established basis for methodological comparison. *Second*, Uniswap operates on the Ethereum blockchain, which hosts the majority of major stablecoins, including both USDT and USDC. *Third*, PancakeSwap’s trading activity includes a non-negligible share of low-liquidity memecoin speculation, which introduces noise that complicates microstructure analysis. *Fourth*, Uniswap is integrated with Dune Analytics, the on-chain data provider used in this thesis.

**On-chain data provider.** Transaction-level blockchain data is required for this thesis, as the key microstructure variables, such as gas prices, block inclusion timing, swap execution, mempool congestion, and liquidity changes, are not available from aggregated data providers. This study therefore relies on Dune Analytics. Dune provides decoded smart contract events extracted directly from the Ethereum blockchain to be queried using a derivative of SQL. Dune’s granularity and reproducibility of queries make it well-suited for academic research, and its native integration with Uniswap v3 pool data facilitates the construction of variables and aggregation of inputs.

#### 4.1.2 Selected Stablecoin Pair and Pool

Since this thesis examines microstructure on both centralized and decentralized exchanges, the selected asset pair must be actively traded on both venues. The most direct proxy for stablecoin peg deviation would be a USD/USDT pair, as Tether is the largest stablecoin by market capitalization at approximately \$184B (DefiLlama, n.d.). However, fiat-crypto pairs are not traded on decentralized exchanges, which are crypto-only by design. The next best option is therefore USDC/USDT, as USD Coin, the second largest stablecoin at approximately \$79B (DefiLlama, n.d.), can replace USD. The market capitalization of both coins is also not irrelevant, as larger stablecoins are backed by deeper asset pools, which in turn supports peg stability. The USDC/USDT pair therefore represents the closest available proxy for a fiat-based stablecoin exchange rate in a crypto-crypto setting.

On Uniswap, four pools exist for the USDC/USDT pair. Three of these operate on the Uniswap v4 protocol, which launched recently and has not yet completed the full transfer of liquidity from v3. The remaining v3 pool has the highest total value locked (TVL) among v3

pools at \$26.7M, with a 30-day trading volume of \$525.6M as of March 24th, 2026 (Uniswap Labs, n.d.-b). Although the v4 pools have attracted higher recent trading volume following their launch, the v4 protocol is too recent to provide a secure base for the analysis, as liquidity has not yet transferred. The v3 pool is therefore selected as the primary DEX data source.

## 4.2 Data Collection Procedures

This section describes the data collection procedures for the three data sources used in this thesis: the Binance CEX, the Uniswap v3 DEX, and Dune Analytics for the Ethereum blockchain. Each source required a different collection approach due to differences in data availability, data types, and underlying querying protocols. CEX data is collected partly from a centralized historical archive and partly through a custom-built real-time system. DEX data is queried via a decentralized subgraph API. On-chain metrics and transaction records are extracted directly from the Ethereum blockchain using SQL queries on Dune Analytics.

All data is collected at, or aggregated to, five-minute intervals aligned to UTC clock boundaries, ensuring consistent timestamps across all sources. The choice of five-minute intervals follows established convention in high-frequency volatility research. Andersen and Bollerslev (1997, 1998) demonstrate that five-minute sampling of intraday return series provides accurate volatility estimates while remaining robust to microstructure noise. The inclusion of microstructure effects presents a well-known trade-off in which higher sampling frequencies introduce noise, while lower frequencies fail to capture intraday variation (Andersen & Bollerslev, 1997, 1998a, 1998b). This standard has carried over into the cryptocurrency volatility literature, where five-minute intervals are widely used for volatility forecasting and modeling on both centralized exchanges and decentralized protocols (Brini & Lenz, 2024; Cao et al., 2009; Hansen et al., 2021). The five-minute frequency is therefore the appropriate choice for this study.

### 4.2.1 CEX Data (Klines and Order Book Reconstruction)

**Kline data.** Historical candlestick data for the USDC/USDT spot pair on Binance is available directly from Binance’s public data repository<sup>1</sup>. Pre-computed five-minute candlesticks are downloaded as CSV files and contain, for each interval, the opening, high, low, and closing price; total trading volume in the base asset; total trading volume in the

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<sup>1</sup>Binance Data Vision: <https://data.binance.vision/>

quote asset; the number of individual trades; and the taker-initiated buy volume in both base and quote assets. The taker buy volumes are useful for differentiating buyer-initiated from seller-initiated transactions.

**Order book data.** Unlike kline data, Binance does not provide historical limit order book snapshots. As such, the collection of order book data requires the construction of a local live copy of the limit order book, of which five-minute snapshots are taken.

The reconstruction follows the official Binance procedure for managing a local order book<sup>2</sup>. The system connects to Binance’s differential depth WebSocket stream, which broadcasts incremental updates to the order book at 1000 ms intervals<sup>3</sup>. On startup, the system buffers a few updates to initialize a metric for how to evaluate the timing of incoming updates. Each push of updates through the WebSocket contains a range of changes to the order book, with the first change being  $U$  and the last change being  $u$ . After buffering updates, the program fetches a full REST snapshot of the current order book at a depth of 1000 price levels per side and initializes a local copy. The REST snapshot contains a `lastUpdateID`, which is set as the starting point for the order book, i.e., any updates with  $u \leq \text{lastUpdateID}$  are ignored, and the first valid event has  $U \leq \text{lastUpdateID} \leq u$ . For example, assume two events with  $[U, u]_1 = [5, 7]$ , and  $[U, u]_2 = [8, 10]$  with a `lastUpdateID` of 9. Then the first event  $[U, u]_1$  is ignored, since  $7 \leq 9$ , and for the second event the update with ID 10 is kept, becoming the bridge from which the WebSocket stream updates the local order book. Subsequent WebSocket messages  $u+1$  are incremental updates to the order book, which add, modify, or remove price levels based on recent transactions. If a gap is detected, or if the connection is lost for any reason, the program discards the local book and automatically attempts to reconnect every five seconds, re-initializing from a fresh REST snapshot upon reconnection. This ensures that the local order book is always consistent with the exchange’s actual state.

Every second, the following metrics are computed from the live order book and stored in memory. The best bid and best ask are the highest buy and lowest sell prices currently in the book. The bid-ask spread is their difference. The *midprice* is their arithmetic mean. Bid depth and ask depth are computed as price multiplied by quantity, denominated in USD, of

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<sup>2</sup>Binance WebSocket Streams documentation: <https://developers.binance.com/docs/binance-spot-api-docs/web-socket-streams#how-to-manage-a-local-order-book-correctly>

<sup>3</sup>Binance Diff-Depth Stream: <https://developers.binance.com/docs/binance-spot-api-docs/web-socket-streams#diff-depth-stream>

the top 20 price levels on each side of the book, following the depth formulation introduced in Section 3.1.1. Finally, *order book imbalance* is defined as:

$$\text{imbalance} = \frac{\text{bid depth} - \text{ask depth}}{\text{bid depth} + \text{ask depth}}.$$

which takes values in  $[-1, 1]$ , with positive values indicating excess buying pressure and negative values indicating excess selling pressure, as discussed in Section 3.1.2.

Every five minutes, aligned to UTC clock boundaries (00:00, 00:05, ...), approximately 300 one-second observations are aggregated into a single record and appended to a daily CSV file. An entry contains all metrics discussed above aggregated over the five-minute window, along with the count of valid one-second observations. The standard deviation of the spread and imbalance are included as additional measures of within-window variability.

#### 4.2.2 DEX Data (Swaps, Liquidity, Mint/Burn Events)

DEX data is collected from the Uniswap v3 subgraph hosted on The Graph<sup>4</sup>, a decentralized indexing protocol that decodes and indexes Ethereum smart contract events into a queryable GraphQL API. Unlike the CEX, where real-time connection was required, the subgraph stores historical swap data permanently, while liquidity states, liquidity flow, and liquidity distribution require being queried at each five-minute interval. The pool queried is the USDC/USDT 0.01% fee pool with address

`0x3416cf6c708da44db2624d63ea0aaef7113527c6`.

Data is collected across three separate query types, each targeting a different aspect of pool activity, and saved to separate daily CSV files.

**Swap data and DEX klines.** Individual swap transactions are fetched for each calendar day using a paginated GraphQL query. As the subgraph API returns a maximum of 1,000 records per request, days with more than 1,000 swaps require cursor-based pagination: Executing the code returns the first 1,000 records, after which the query is run for another time using the 1,000th swap’s ID as the starting point, until a batch of fewer than 1,000 results is retrieved, signaling that all records have been queried. Each swap record contains the transaction timestamp, the signed token amounts exchanged (USDC `amount0`, USDT `amount1`), the USD value of the swap, the square root price after execution

<sup>4</sup>Uniswap v3 subgraph ID: `5zvR82QoaXYFYDEKLZ9t6v9adgnptxYpKpSbxtgVENFV`

(`sqrtPriceX96`), the resulting tick, and the sender and recipient addresses.

The raw swap records are saved to CSV, and five-minute klines are constructed from them by aggregating all swaps whose timestamp falls within a given UTC-aligned window. Each kline contains the opening, high, low, and closing price derived from the post-swap square root prices, the total USD volume, the number of swaps, a directional order flow imbalance measure (the net `amount0` divided by total absolute `amount0`), as well as a count and volume metrics for large trades (defined as swaps exceeding \$100,000 in USD value), which may capture institutional order flow or arbitrage activity.

**Pool state (liquidity state).** Every five minutes the current state of the pool is queried. The response includes the active liquidity parameter  $L$ , the current tick and square root price `sqrtPriceX96`, the derived token prices (`token0Price`, `token1Price`), and the total value locked in each token and in USD. The price is recomputed from the `sqrtPriceX96` value returned by the subgraph, which stores price as  $\sqrt{p} \times 2^{96}$  by inverting the operation. Since both USDC and USDT carry six decimal places, no decimal adjustment is required. These periodic snapshots capture the evolution of pool depth and price over time. The pool price is used to construct the return series because, unlike the klines, the pool price is recorded every five minutes regardless of whether a swap occurred or not. Thus the data is complete and does not require manual imputation.

**Mint and burn events (liquidity flow).** This query fetches all mint and burn events that occurred since the previous window. Mints represent liquidity additions by LPs, while burns represent liquidity withdrawals (these are LP events, which are not the same as the minting and burning of stablecoin supply discussed in Chapter 2). Each event record contains the timestamp, the liquidity amount added or removed, the token amounts deposited or withdrawn, the USD value, and the lower and upper tick bounds of the affected position. The five-minute aggregation records the total number of mints and burns, the summed liquidity and token amounts for each, and the net liquidity change over the window, which captures whether the pool is experiencing net inflows or outflows.

**Tick data (liquidity distribution).** At the same five-minute intervals, the 20 ticks immediately surrounding the current tick, i.e., ticks in the discrete interval  $[10, -10]$ , are queried. In the Uniswap v3 protocol, each initialized tick stores two liquidity values that together describe how much liquidity is available in the neighborhood of the current price (Adams et al., 2021; Jeiwan, n.d.). `liquidityGross` is the total absolute liquidity

referencing that tick, i.e., all LP positions that use this tick as either their lower or upper boundary serving as an indicator of how much liquidity is concentrated near this price level. `liquidityNet` records how much active liquidity changes when the price crosses through that tick. When the price moves upward through a tick, `liquidityNet` is added to the pool's active liquidity: a positive value at a tick means entering a new LP position's range (liquidity is added), while a negative value means exiting an LP position's range (liquidity is removed). The sign is reversed when price moves downward. From the queried ticks, three aggregate metrics are constructed: `total_liq_gross`, the sum of `liquidityGross` across all ticks in the window, which measures total liquidity concentration near the current price; `net_liq_above`, the sum of `liquidityNet` for ticks above the current tick, which indicates how much liquidity would be activated or deactivated if the price moves upward; and `net_liq_below`, the equivalent for ticks below, capturing liquidity changes on a potential downward move. Together these metrics provide a local picture of the liquidity distribution around the current price, which is directly relevant to predicting price impact and volatility, as discussed in Section 3.2.

#### 4.2.3 On-Chain Metrics (Transfers, Gas, Congestion)

On-chain metrics are collected using Dune Analytics, a platform that provides SQL access to decoded Ethereum blockchain data, including transaction records, block headers, and decoded smart contract events<sup>5</sup>. Unlike the CEX or DEX data collection methods, Dune queries run against a fully indexed historical database, allowing the entire sample period to be collected retrospectively in a single query execution. All six queries, used to cover distinct on-chain phenomena, produce five-minute aggregates.

**Whale transfers (Query 1).** This query identifies large USDC and USDT token transfers on Ethereum above a configurable USD threshold. For each transfer, it records the block timestamp, sender and receiver addresses, token amount, and USD value, and tags each transfer with a flow direction, i.e., whether the recipient, sender, both, or none is an exchange, by joining against a table of known centralized exchange addresses. Large transfers between wallets and exchanges are a proxy for institutional repositioning and are expected to precede or coincide with periods of elevated volatility.

**CEX inflows and outflows (Query 2).** This query aggregates CEX inflows and outflows

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<sup>5</sup>Dune Analytics: <https://dune.com>

for USDC and USDT using Dune’s pre-labelled `cex.flows` table, which covers all token movements into and out of known exchange addresses. For each five-minute window, it computes the total number of transfers, total and average USD amounts, and the net flow (inflows minus outflows), broken down by exchange name and flow type. CEX flow imbalances capture the directional pressure of large token movements and serve as a complement to the on-exchange order book imbalance collected from Binance and Query 1.

**Gas prices (Query 3).** This query constructs a five-minute gas price series by joining block-level and transaction-level data. From the block table it extracts the average, minimum, maximum, and median base fee per gas in Gwei for each window, covering all blocks mined within that interval (approximately 25 blocks at Ethereum’s current block time of around 12 seconds). Under the EIP-1559 fee mechanism, the base fee is a protocol-set floor price for inclusion, adjusted algorithmically based on the fullness of the previous block (Buterin et al., 2019). From the transaction table it extracts the 10th, 50th, and 80th percentiles of total gas price and of the priority fee for EIP-1559 transactions, computed over all transactions in the five-minute window rather than a single minute, making the percentile estimates substantially more stable. An approximate effective gas price (base fee plus median priority fee) is also computed as a summary measure.

The priority fee columns (`dune_gas_tip_p10/p50/p80_gwei`) are not directly available from Dune’s `ethereum.transactions` table and are instead approximated in preprocessing as the difference between the corresponding total gas price percentile and the block-level base fee:

$$\begin{aligned} \text{dune\_gas\_tip\_p}\{10,50,80\}\_\text{gwei} &\approx \text{dune\_gas\_price\_p}\{10,50,80\}\_\text{gwei} \\ &\quad - \text{dune\_gas\_base\_fee\_}\{\text{min,avg,max}\}\_\text{gwei}. \end{aligned}$$

**Block utilization (Query 4).** This query measures how full each Ethereum block is. The key metric is the gas utilization ratio: gas used divided by the gas limit. The query records the average, median, and maximum utilization per window, covering all blocks within the interval, as well as the percentage of blocks exceeding 50% and 80% of capacity. These thresholds were economically established in Chapter 2: utilization above 50% begins to exert upward pressure on the base fee, and utilization above 80% signals near-full blocks in which competition for inclusion is intense. The query additionally records average block

gas usage and average block size in bytes, which capture broader network load beyond standard transaction activity.

**Congestion composite score (Query 5).** This query combines several block- and transaction-level signals into a composite congestion proxy at the five-minute level. It assembles the average block fill ratio, the percentage change in the base fee relative to the previous five-minute window, and the priority fee spread between the 80th and 10th percentiles of EIP-1559 transactions, combining them into a single score bounded between 0 and 1:

$$\text{congestion score} = \min\left(1, 0.4 \times \text{fill ratio} + 0.3 \times \mathbf{1}_{\Delta \text{base fee} > 0} + 0.3 \times \frac{\min(\text{priority spread}, 20)}{20}\right).$$

*It is important to note that while the components of this formula are grounded in literature, such as Huberman et al. (2021), the weights are assigned heuristically and the congestion score is a composite proxy. The weighing is not based on a theoretical model.*

The fill ratio component captures current network load; the base fee change component captures whether congestion is increasing or decreasing relative to the previous window; and the priority fee spread captures the degree of competition for block inclusion, as a wider spread indicates that some users are bidding substantially more than others to gain priority (see the gas auction mechanism discussed in Section 3.3.2). Using five-minute windows rather than one-minute windows improves the stability of the percentile-based spread estimate.

The mempool priority fee columns (`dune_mempool_tip_median/p80/spread_gwei`) suffer from the same data availability issue as Query 3 and are dropped entirely, with the equivalent tip approximations from Query 3 used in their place.

**Protocol-level mints and burns (Query 6).** This query captures true supply-side events, such as the issuance and redemption of stablecoins at the protocol level, as opposed to the LP-level mint and burn events in the Uniswap pool. For USDC, these are the `Mint` and `Burn` events emitted by Circle’s FiatToken contract. For USDT, the equivalent events are `Issue` (minting) and `Redeem` (burning), along with `DestroyedBlackFunds` events representing forced burning from blacklisted addresses. This rare event occurs, for example, when legal action is taken against sanctioned entities or fraud cases (Bitget, 2026; Tether Operations Limited, n.d.-b, Art. 8). For each five-minute window, the query counts events

and aggregates the token amounts issued or redeemed per token and event type and computes a running cumulative net supply change across all windows. As discussed in Chapter 2, large issuance events signal institutional demand for stablecoins, while large redemptions may reflect capital outflows or stress, both of which can affect peg stability and short-term volatility.

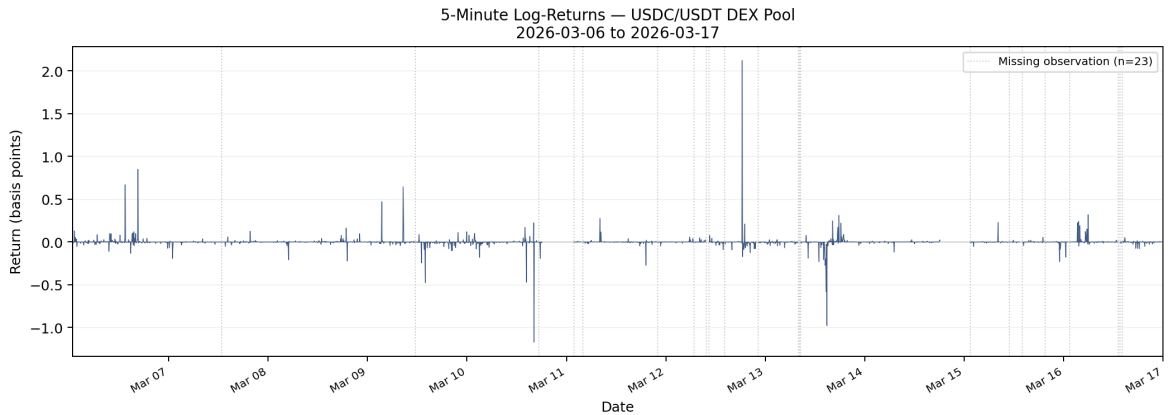
### 4.3 Data Cleaning

The first complete observation is recorded at 2026-03-06 00:50:00, and as such the return series begins at 2026-03-06 00:55:00. Returns are computed as log returns scaled to basis points:

$$r_t \equiv 10,000 \times \ln\left(\frac{p_t}{p_{t-1}}\right).$$

and is visualized in Figure 4.1

Figure 4.1: Five-minute log returns of the USDC/USDT DEX pool, March 2026, with gap boundaries marked.



Scaling to basis points is appropriate given that price deviations in the USDC/USDT pair are already on that order of magnitude. Duplicate timestamps arising from merging data sources were removed by retaining the first observation per five-minute window.

A total of 63 five-minute windows were absent due to data collection interruptions affecting the DEX feed. The gaps consist of 16 isolated single-window interruptions, one gap of two consecutive windows, one of three, two of four, and two of six, as well as one extended interruption of 21 consecutive windows (105 minutes) between 2026-03-11 00:00:00 and 2026-03-11 01:40:00. For each gap, the return observation immediately

following the resumption of data is additionally excluded, as it would otherwise span multiple five-minute periods and produce an inflated return that the GARCH model would incorrectly treat as a single-period shock. For example, since data collection was interrupted for one observation at 2026-03-07 12:45:00, the return series resumes at 2026-03-07 12:55:00 rather than 12:50:00.

Additionally, there were 3 consecutive data gaps amounting to 152 five-minute windows in which the CEX order book data were removed entirely due to collection interruptions in real-time order book recording. The return values for the immediate observation after the removed window were dropped for the same reason as above, although one of those has already been dropped as it followed a gap in the DEX feed. Retaining these windows would introduce regime inconsistency into the model inputs, as the set of input features would differ structurally between windows where order book variables are observed and windows where they are absent, which would cause the model to learn incorrect patterns driven by data (un)availability instead of basing prediction on underlying market dynamics.

Ultimately, 2942 complete observations and 2919 entries for the return series have been collected between 2026-03-06 00:50:00 and 2026-03-17 00:00:00.

## 5. Volatility Modeling

This chapter introduces the econometric foundation for volatility modeling. The objective of the thesis is to build a deep learning neural network that adjusts the parameters of an econometric model to predict volatility based on market microstructure variables. The underlying econometric model proposed by Bollerslev (1986), the Generalized Autoregressive Conditional Heteroscedasticity (GARCH) model, therefore serves as the benchmark against which the deep learning hybrid model is evaluated. Section 5.1 covers volatility from the perspective of financial time series. Section 5.2 introduces the GARCH model and its use in volatility prediction. Section 5.3 justifies the use of the GARCH model as the benchmark for stablecoin volatility prediction.

### 5.1 Definition and Characteristics of Volatility

Volatility in financial time series refers to the conditional variance  $\sigma_t^2 = \text{Var}(r_t \mid F_{t-1})$  of an asset's return series  $(r_t)_{t \in \mathbb{N}}$  (Tsay, 2010, p. 109). More formally, consider a return series  $(r_t)_{t \leq T}$  with conditional mean  $\mu_t = E(r_t \mid F_{t-1})$  and conditional variance  $\sigma_t^2 = \text{Var}(r_t \mid F_{t-1})$ , where  $F_{t-1}$  denotes the information set available at time  $t - 1$  (Tsay, 2010, p. 111).

Returns are assumed to follow an autoregressive moving average (ARMA) model of order  $p$  and  $q$ :

$$r_t = \mu_t + a_t, \quad \mu_t = \sum_{i=1}^p \phi_i y_{t-i} - \sum_{i=1}^q \theta_i a_{t-i}, \quad y_t = r_t - \phi_0 - \sum_{i=1}^k \beta_i x_{it}.$$

where  $a_t$  is referred to as the *innovation* or *shock* at time  $t$ . It follows that:

$$\sigma_t^2 = \text{Var}(r_t \mid F_{t-1}) = \text{Var}(\mu_t + a_t \mid F_{t-1}) = \text{Var}(a_t \mid F_{t-1}).$$

Conditional heteroscedasticity models, such as GARCH, are concerned with how  $\sigma_t^2$  evolves over time.

Empirically, financial volatility exhibits several well-documented characteristics (Tsay, 2010, pp. 110, 111). *First*, volatility is not directly observable, as it represents the

dispersion of shocks around the conditional mean and requires multiple return observations to estimate reliably. In the context of stablecoins, estimation is simplified by assuming that the mean return  $\mu_t$  is zero, i.e., that prices revert to parity in the long run, so that the innovation  $a_t$  can be approximated by the return  $r_t \approx a_t$ , removing the need to separately estimate the conditional mean. *Second*, we observe volatility clustering, which refers to the fact that periods of high volatility tend to be followed by further elevated volatility, and periods of calm tend to persist. This implies that current volatility depends on its own recent history. *Third*, volatility evolves continuously over time, meaning that jumps are rare. *Fourth*, volatility responds asymmetrically to price increases and price decreases, which we call the leverage effect (which standard GARCH does not capture and which is noted here as a limitation of the benchmark model). *Fifth*, volatility is mean-reverting and does not diverge to infinity.

## 5.2 The GARCH Framework

The GARCH model, proposed by Bollerslev (1986), extends the Autoregressive Conditional Heteroscedasticity (ARCH) model of Engle (1982). The ARCH( $m$ ) model specifies conditional variance as  $\sigma_t^2 = \alpha_0 + \sum_{i=1}^m \alpha_i a_{t-i}^2$ , with  $a_t = \sigma_t \epsilon_t$ . A practical limitation of ARCH is that capturing persistent volatility requires a large lag order  $m$ , making the model parameter-heavy and prone to overfitting, which forces large shocks to inflate forecasts over many periods even after volatility has stabilized (Tsay, 2010, p. 119). The GARCH model addresses this by augmenting the shock terms with a lagged conditional variance component, parsimoniously capturing volatility persistence.

### 5.2.1 Model Specification

The GARCH( $m, s$ ) model is specified as:

$$a_t = \sigma_t \epsilon_t,$$

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^m \alpha_i a_{t-i}^2 + \sum_{j=1}^s \beta_j \sigma_{t-j}^2.$$

where  $(\epsilon_t)_{t \in \mathbb{N}}$  is a sequence of iid standard normal random variables,  $\alpha_0 > 0$ ,  $\alpha_i \geq 0$  for all  $i \leq m$ ,  $\beta_j \geq 0 \forall j \leq s$ , and  $\sum_{i=1}^{\max(m,s)} (\alpha_i + \beta_i) < 1$ , which ensures that the unconditional variance of  $a_t$  is finite and that the process is covariance-stationary (Tsay, 2010, p. 132).

For a GARCH(1,1) model,  $\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2$ , the one-step forecast from origin  $h$  is:

$$\sigma_{h+1}^2 = \sigma_h^2(1) = \alpha_0 + \alpha_1 a_h^2 + \beta_1 \sigma_h^2.$$

At the five-minute frequency used in this thesis, the one-step forecast determines the next five-minute predicted volatility. One can show through substitution of  $a_t^2 = \sigma_t^2 \epsilon_t^2$  that for an  $\ell$ -step forecast from origin  $h$ , the  $\ell^{th}$  period volatility is given by

$$\begin{aligned} \sigma_h^2(\ell) &= \alpha_0 + (\alpha_1 + \beta_1) \sigma_h^2(\ell - 1), & \ell > 1, \\ \Leftrightarrow \sigma_h^2(\ell) &= \frac{\alpha_0 [1 - (\alpha_1 + \beta_1)^{\ell-1}]}{1 - \alpha_1 - \beta_1} + (\alpha_1 + \beta_1)^{\ell-1} \sigma_h^2(1), \\ \Leftrightarrow \sigma_h^2(\ell) &\rightarrow \frac{\alpha_0}{1 - \alpha_1 - \beta_1}, & \text{as } \ell \rightarrow \infty. \end{aligned}$$

which is the unconditional variance of  $a_t$  (Tsay, 2010, p. 133).

### 5.2.2 Interpretation of Parameters

The GARCH(1,1) model has two parameters governing the dynamics of conditional variance. Like ARMA models used for return series (Tsay, 2010, pp. 64-71), the GARCH model employs both a short-term and long-term memory component to model the dynamics of volatility.

The ARCH parameter  $\alpha$  measures the short-term impact shocks of volatility. In other words,  $\alpha$  describes how a sudden change in prices affects the price dynamics for the following periods. A high  $\alpha$  would imply that prices are sensitive to shocks and tend to experience high volatility in the short term. We call this short-term memory, since the effect of past shocks fades after a few periods.

The ARCH parameter  $\alpha_1$  governs the short-term impact of shocks on volatility. It captures how a sudden change in returns affects the conditional variance in the periods immediately following. A large  $\alpha_1$  implies that volatility is highly reactive to recent shocks. Generally, the ARCH parameter describes short-term volatility that dissipates relatively quickly, which is what we call short-term memory.

The GARCH parameter  $\beta_1$  governs the persistence of past conditional variance. A large  $\beta_1$  indicates that volatility shocks decay slowly, so that elevated volatility today implies

elevated volatility for an extended subsequent period. We call this long-term memory, since past volatility continues to influence present volatility.

The sum  $\alpha + \beta$  thus reflects the overall persistence of volatility. Values close to 1 indicate that volatility is highly persistent, as by construction we require  $\alpha + \beta < 1$ . Thus, the GARCH allows to capture volatility clustering, where periods of high volatility are followed by high volatility, and smooths out shocks through the GARCH parameter.

The sum  $\alpha_1 + \beta_1$  thus reflects overall volatility persistence. Values close to 1 indicate highly persistent volatility dynamics, consistent with the empirical observation of volatility clustering. The stationarity constraint  $\alpha_1 + \beta_1 < 1$  ensures that the process reverts to its unconditional variance in the long run.

### 5.3 Justification as Benchmark Model

GARCH-type models are the standard econometric tool for volatility modeling and forecasting due to their ability to capture the empirical characteristics of financial return series. They have been extensively applied to traditional asset classes, including foreign exchange rates (Dritsaki, 2019; Johnston & Scott, 2000; Mundaca, 1991; Yoon & Lee, 2008) and equities (Alberg et al., 2008; Boako et al., 2015; Worthington & Higgs, 2004), as well as to cryptocurrencies (Chu et al., 2017; Dyhrberg, 2016; Ngunyi et al., 2019) and stablecoins specifically (Chen & Chang, 2022; Lee, 2025). A growing literature also combines machine learning methods with GARCH frameworks (Amirshahi & Lahmiri, 2023; Beelen, 2025; García-Medina & Aguayo-Moreno, 2024; Michańków et al., 2023; Mostafa et al., 2021).

As such, the GARCH model represents a natural and well-validated benchmark. In this thesis, the GARCH(1,1) specification is used to evaluate whether incorporating non-linearity and interactions among heterogeneous market microstructure variables, through a deep learning hybrid model, improves upon standard volatility prediction.

## 6. Hybrid Deep Learning GARCH Model

This chapter develops the hybrid deep learning GARCH model and discusses the results and limitations of the application of the hybrid GARCH on the collected data. Section 6.1 describes the model architecture and the design of the output constraints. Section 6.2 covers the training procedure, including the loss function, optimizer, and the progression of training runs. Section 6.3 presents the benchmark GARCH, the hybrid model, permutation feature importance, and forecasting results. Section 6.4 discusses the limitations of the model.

### 6.1 Model Architecture

The model architecture must satisfy two requirements simultaneously. *First*, it must enforce the stationarity conditions of the GARCH, ensuring that the time-varying parameters produced by the network are valid inputs to the variance recursion at every subsequent time step. *Second*, the architecture must be appropriate for the available data, as the sample is limited in size and, as documented in Section 4.3, contains 26 gaps (23 due to DEX interruptions + 3 due to CEX interruptions) of varying length, introduced by data collection interruptions, which constrain the class of models that can be applied without corrupting the input sequence.

#### 6.1.1 Network Structure and Design Choices

The deep learning component is a feedforward neural network with two hidden layers of 32 neurons each. Formally, the network maps the 41-dimensional input vector (see Table 6.1) at time  $t$  through the following sequence of transformations:

$$\begin{aligned} h_1 &= \text{ReLU}(W_1 x_t + b_1), & W_1 &\in \mathbb{R}^{32 \times 41}, \\ h_2 &= \text{ReLU}(W_2 h_1 + b_2), & W_2 &\in \mathbb{R}^{32 \times 32}, \\ z_t &= W_3 h_2 + b_3, & W_3 &\in \mathbb{R}^{2 \times 32}. \end{aligned}$$

producing two raw output logits  $z_t = (z_{1,t}, z_{2,t})$  per time step. The activation function ReLU, which is defined as  $\text{ReLU}(x) = \max(0, x)$ , sets any negative value to zero while passing positive ones through. This introduces the nonlinearity required to capture interactions between microstructure variables that a linear model could not represent, while remaining computationally simple and well-suited to small datasets.

The network takes 41 concurrent microstructure features as input at each time step  $t$ . All inputs are grouped by data source and summarized in Table 6.1 The input n-tuple for a

Table 6.1: Microstructure input features to the GARCH-NN model, grouped by data source.

| Category         | Features                                                                                                                                                 |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| CEX trading      | cex_volume_usdc, cex_n_trades, cex_taker_buy_sell_ratio, cex_ob_imbalance_mean, cex_ob_bid_depth_mean, cex_ob_ask_depth_mean, cex_ob_spread_mean         |
| DEX swaps        | dex_klines_volume_usd, dex_klines_n_swaps, dex_klines_imbalance, dex_klines_large_trades_count, dex_klines_large_trades_usd                              |
| DEX pool         | dex_pool_liquidity, dex_pool_tvl_usd                                                                                                                     |
| DEX ticks        | dex_ticks_total_liq_gross, dex_ticks_net_liq_above, dex_ticks_net_liq_below, dex_ticks_n_active                                                          |
| DEX LP           | dex_lp_net_liq_change, dex_lp_n_mints, dex_lp_n_burns                                                                                                    |
| Gas              | dune_gas_base_fee_gwei, dune_gas_tip_p50_gwei, dune_gas_tip_p80_gwei, dune_gas_effective_gwei                                                            |
| Blocks & mempool | dune_mempool_congestion_score, dune_block_utilization, dune_block_pct_near_full, dune_mempool_base_fee_change                                            |
| Protocol supply  | dune_supply_usdc_burn_total_token_amount, dune_supply_usdc_mint_total_token_amount, dune_supply_usdt_burn_blacklist_total_token_amount                   |
| CEX flows        | dune_flows_usdc_inflow_total_usd, dune_flows_usdc_outflow_total_usd, dune_flows_usdt_inflow_total_usd, dune_flows_usdt_outflow_total_usd                 |
| Whale transfers  | dune_whale_usdc_cex_inflow_total_usd, dune_whale_usdc_cex_outflow_total_usd, dune_whale_usdt_cex_inflow_total_usd, dune_whale_usdt_cex_outflow_total_usd |
| Derived          | dex_cex_price_spread                                                                                                                                     |

return  $r_t$  in period  $t$  only contains microstructure values of the past five minutes, i.e., of the same observation, conditioning the parameters only on the most recent data.

The decision to use only immediate inputs, rather than a sequence of lagged observations, is traced back to the gap structure of the data. A potential alternative would have been to aggregate market conditions over a longer lookback window, for example, 30

minutes, and feed this as an aggregated state input to the feedforward network, partially replicating the role that a recurrent hidden state would play in an LSTM. However, given that gaps occur throughout the sample, any fixed-length lookback window would regularly span a gap boundary. For example, if a gap occurs at 10:00:00, then any lookback between 10:05:00 and 10:30:00 would incorporate one less observation, which would not faithfully represent the actual number of transfers, burns, and mints, or changes in depth during that window, rendering any training with those corrupted windows invalid. The same logic applies to LSTM models more directly. Their recurrent hidden state is built up sequentially, and as such a gap in the data would cause the hidden state passed forward to incorrectly represent the past, corrupting all subsequent predictions until the sequence resets.

### 6.1.2 Output Constraints and Parameter Bounds

As established in Section 5.2.1, the GARCH variance recursion requires that  $\alpha_t > 0$ ,  $\beta_t > 0$ , and  $\alpha_t + \beta_t < 1$  at every time step to ensure that the conditional variance remains positive and that the process is covariance-stationary. For that reason, raw network outputs, which are unconstrained real-valued logits, must be transformed before being passed into the recursion.

The natural first approach is softmax. For a vector of raw logits  $z = (z_1, z_2)$ , the softmax function is defined as:

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}.$$

which maps any real-valued vector to a probability simplex, i.e., all outputs are strictly positive and sum to exactly one. Scaling by  $1 - \varepsilon$  with  $\varepsilon = 0.001$  therefore gives:

$$(\alpha_t, \beta_t) = (1 - \varepsilon) \times \text{softmax}(z_{1,t}, z_{2,t}).$$

This guarantees  $\alpha_t, \beta_t > 0$  and  $\alpha_t + \beta_t = 0.999 < 1$  simultaneously, satisfying all stationarity conditions in a single operation with no additional penalty terms. The softmax constraint is therefore theoretically elegant and was the natural starting point.

However, the softmax construction introduces a structural problem. The constraint that  $\alpha_t + \beta_t = 0.999$  implies that increasing one factor decreases the other by the same amount. The network therefore does not learn separate microstructure conditions for shock reactivity and volatility persistence independently but instead learns a single allocation

between them. As further discussed in Section 6.3, this zero-sum structure interacts badly with the character of the USDC/USDT sample. Ultimately, an alternative constraint design fits the sample better.

The alternative uses the logistic sigmoid function, defined as:

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

which maps any real number to the interval  $(0, 1)$  independently. Unlike softmax, the sigmoid is applied to each output separately, so  $\alpha_t$  and  $\beta_t$  each have their own unconstrained pathway through the network. Each output is then scaled by an upper bound:

$$\alpha_t = a \times \sigma(z_{1,t}), \quad \beta_t = b \times \sigma(z_{2,t}).$$

The bounds  $a$  and  $b$  are set according to parameter ranges that are well-established in the GARCH literature. Across asset classes and frequencies,  $\alpha$  consistently falls in the range 0.05–0.25 and  $\beta$  in the range 0.70–0.93, with  $\alpha + \beta$  typically between 0.95 and 0.99, reflecting highly persistent volatility dynamics (Amirshahi & Lahmiri, 2023; Andersen & Bollerslev, 1997, 1998a; Beelen, 2025; Bollerslev, 1986; Chu et al., 2017; Dritsaki, 2019; Dyhrberg, 2016; García-Medina & Aguayo-Moreno, 2024). Table 6.2 summarizes these estimates across the papers most relevant to this thesis.

Based on these ranges, the bounds are set to  $a = 0.10$  and  $b = 0.90$ , placing the midpoint of each parameter’s range at approximately the literature consensus. A useful consequence of this design is that at zero initialization, where  $\sigma(0) = 0.5$ , the network recovers  $\alpha_t = 0.05$  and  $\beta_t = 0.45$  as a prior, well within the empirically plausible region. Since the sigmoid output lies strictly in  $(0, 1)$ , both parameters are positive by construction. The stationarity constraint  $\alpha_t + \beta_t < 1$  is enforced through a differentiable post-hoc rescaling: whenever  $\alpha_t + \beta_t \geq 1$  at any time step, both parameters are multiplied by  $0.999 / (\alpha_t + \beta_t)$ , reducing their sum to just below unity while preserving their relative magnitude. In practice this correction is rarely triggered given the chosen bounds.

## 6.2 Training Procedure

The GARCH recursion is sequential, meaning that  $\sigma_t^2$  depends on  $\sigma_{t-1}^2$  and  $a_{t-1}^2$  at every step. The entire training window is therefore passed as a single batch per epoch, with

Table 6.2: GARCH(1,1) parameter estimates from selected papers.

| Paper                                | Asset                  | $\hat{\alpha}$ | $\hat{\beta}$ | $\hat{\alpha} + \hat{\beta}$ |
|--------------------------------------|------------------------|----------------|---------------|------------------------------|
| Bollerslev (1986)                    | Macro / interest rates | 0.10–0.20      | 0.70–0.85     | 0.85–0.98                    |
| Andersen & Bollerslev (1997/1998)    | FX (intraday/daily)    | 0.05–0.15      | 0.85–0.93     | 0.95–0.99                    |
| Dyhrberg (2016)                      | Bitcoin                | 0.20           | 0.78          | 0.98                         |
| Chu et al. (2017)                    | Bitcoin                | 0.13           | 0.86          | 0.99                         |
|                                      | Other cryptos          | 0.10–0.25      | 0.70–0.88     | 0.95–0.99                    |
| Dritsaki (2019)                      | FX                     | 0.08–0.15      | 0.80–0.90     | 0.95–0.99                    |
| Amirshahi & Lahmiri (2023)           | Crypto                 | 0.10–0.22      | 0.75–0.88     | 0.95–0.99                    |
| García-Medina & Aguayo-Moreno (2024) | Crypto portfolios      | 0.12–0.20      | 0.80–0.87     | 0.96–0.99                    |
| Beelen (2025)                        | Crypto                 | 0.10–0.18      | 0.82–0.90     | 0.95–0.99                    |

backpropagation tracing through the full recursion. This section describes the loss function and optimizer and documents the progression of training runs that led to the final model.

### 6.2.1 Loss Function and Optimizer

The model is trained by minimizing the negative Gaussian log-likelihood over all valid, non-gap timesteps in the training window:

$$\mathcal{L} = \frac{1}{T^*} \sum_{t \in \mathcal{T}^*} \left[ \ln(\sigma_t^2) + \frac{r_t^2}{\sigma_t^2} \right].$$

where  $\mathcal{T}^*$  denotes the set of non-gap timesteps and  $T^*$  its cardinality. The first term penalizes the model for predicting unnecessarily large variance, while the second penalizes it for predicting variance that is too small to explain the observed return. This is the same criterion used to estimate the benchmark GARCH(1,1), ensuring that any difference in performance between the two models reflects the architecture rather than the objective. At gap boundaries,  $\sigma_t^2$  is reset to the unconditional variance  $\omega/(1 - \alpha_t - \beta_t)$ , and the timestep is excluded from the loss, so no information flows across gaps during training. The model is optimized using Adam (Kingma & Ba, 2014), an adaptive gradient descent algorithm that maintains per-parameter learning rates. Standard gradient descent applies the same learning rate to all parameters, which is problematic here because the microstructure features differ substantially in scale and frequency of informative gradients: gas price variables, for instance, spike rarely, while order flow variables fluctuate every window. Adam addresses this by scaling each parameter’s update by its own gradient history, effectively giving rarely updated parameters larger steps and frequently updated parameters smaller ones, which

accelerates convergence without requiring manual tuning of per-feature learning rates. Adam tracks two quantities for each parameter across epochs: the running average of past gradients, which acts as a momentum term that smooths the update direction, and the running average of past squared gradients, which scales the step size down for parameters that consistently receive large updates and up for those that receive small ones. A learning rate of  $10^{-4}$  and L2 weight decay of  $10^{-4}$  are used, the latter discouraging the network from assigning disproportionate weight to any single feature.

Gradient clipping is additionally applied during each backward pass. Gradient clipping refers to a technique that is used to limit excessively large updates to network weights. If the norm of the gradient vector across all parameters exceeds a threshold of 1.0, the entire gradient is rescaled so that its norm equals exactly 1.0, leaving the direction unchanged while limiting the step size. This prevents occasional large gradient steps, which can occur during early training when the GARCH recursion produces extreme variance estimates, from pushing the parameters into a region from which meaningful downward adjustment is difficult.

### 6.2.2 Training Runs

The final model specification was arrived at through iterative experimentation across nine training runs, adjusting the learning rate, number of epochs, weight decay, and output constraint design based on the observed behavior of each run. Table 6.3 summarizes the configurations and key outcomes. The softmax runs (1 through 5) established that 500

Table 6.3: Configuration and outcomes of the nine training runs. WD denotes L2 weight decay.

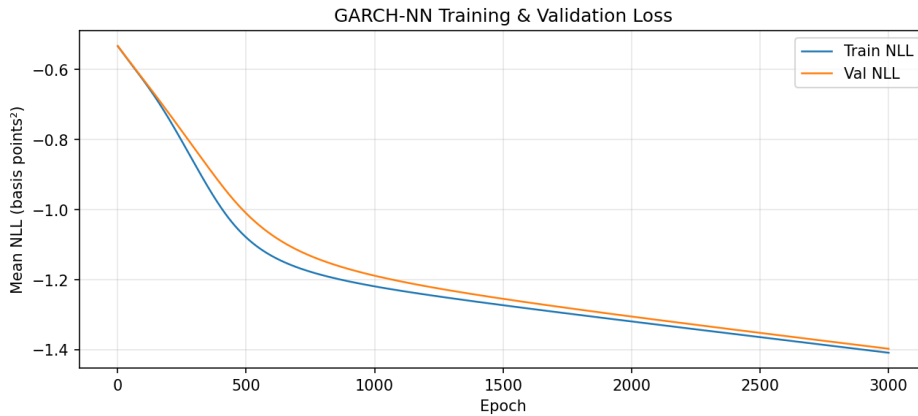
| Run | Epochs | LR   | WD   | Constraint | Convergence | Parameter behavior                               |
|-----|--------|------|------|------------|-------------|--------------------------------------------------|
| 1   | 500    | 1e-3 | 1e-4 | Softmax    | No          | —                                                |
| 2   | 3000   | 1e-3 | 1e-4 | Softmax    | Yes         | $\alpha \approx 0.999$ , unstable $\beta$        |
| 3   | 3000   | 1e-4 | 1e-4 | Softmax    | Yes         | $\alpha \approx 0.999$ , $\beta \approx 0.0002$  |
| 4   | 3000   | 1e-4 | 1e-3 | Softmax    | Yes         | $\alpha \approx 0.999$ , $\beta \approx 0$       |
| 5   | 6000   | 1e-4 | 1e-3 | Softmax    | Yes         | No improvement over run 4                        |
| 6   | 500    | 1e-3 | 1e-4 | Sigmoid    | No          | $\alpha + \beta \approx 0$                       |
| 7   | 3000   | 1e-3 | 1e-4 | Sigmoid    | Yes         | $\alpha \approx \beta \approx 0.001$             |
| 8   | 3000   | 1e-4 | 1e-4 | Sigmoid    | Yes         | $\alpha \approx 0.0016$ , $\beta \approx 0.003$  |
| 9   | 6000   | 1e-4 | 1e-4 | Sigmoid    | Yes         | $\alpha \approx 0.0006$ , $\beta \approx 0.0004$ |

epochs at a learning rate of  $10^{-3}$  was insufficient for convergence. Extending to 3000 epochs in runs 2 and 3 achieved convergence but revealed the zero-sum collapse described

in Section 6.1.2: across all softmax runs,  $\alpha_t$  collapsed to approximately 0.999 while  $\beta_t$  was driven toward zero, regardless of learning rate or weight decay. Adding weight decay in runs 4 and 5 marginally stabilized training but did not change the structural outcome, and extending to 6000 epochs in run 5 produced no improvement over run 4, confirming that the problem was architectural rather than a matter of insufficient training.

Switching to the sigmoid constraint in runs 6 through 9 resolved the zero-sum collapse but introduced the opposite problem: both parameters collapsed toward zero rather than one, with  $\alpha_t + \beta_t \approx 0.002$  across all sigmoid runs. Run 6 again confirmed that 500 epochs at  $10^{-3}$  is insufficient. Run 7 achieved convergence, but at a learning rate of  $10^{-3}$ , the parameters showed minimal variation. Run 8, with a reduced learning rate of  $10^{-4}$  and 3000 epochs, produced the most meaningful parameter variation of all sigmoid runs, with  $\alpha_t \approx 0.0016$  and  $\beta_t \approx 0.003$ , and is selected as the final, albeit degenerated, model. Extending to 6000 epochs in run 9 caused both parameters to collapse further toward zero, suggesting that additional training beyond convergence pushes the network toward a more degenerate solution rather than improving it. Run 8 is therefore the final model, trained for 3000 epochs at a learning rate of  $10^{-4}$  with a weight decay of  $10^{-4}$ . The loss curve for Run 8 is displayed in Figure 6.1.

Figure 6.1: Training and validation loss for Run 8 over 3,000 epochs.



### 6.3 Results and Diagnostics

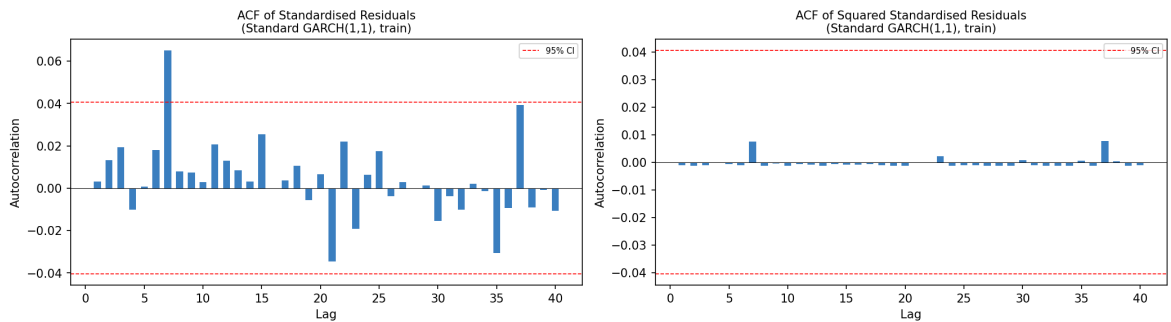
This section presents the empirical results of both the benchmark GARCH(1,1) and the hybrid model, along with the residual diagnostics and permutation feature importance for Run 8. The benchmark results establish whether the data supports GARCH-type modeling in the first place, providing the necessary context for interpreting the hybrid model's behavior.

### 6.3.1 Benchmark GARCH(1,1) Estimation

Before presenting the hybrid model results, it is necessary to discuss what the standard GARCH(1,1) benchmark produces on this data. The model is estimated on the training window of 2,338 valid observations by maximum likelihood, yielding  $\hat{\omega} = 0.3025$ ,  $\hat{\alpha} = 0.9267$ , and  $\hat{\beta} = 0.0733$ . All three estimates are statistically insignificant, with p-values of 0.156, 0.488, and 0.366, respectively, and confidence intervals that include zero and, for  $\hat{\alpha}$ , extend into negative territory. The persistence estimate  $\hat{\alpha} + \hat{\beta} \approx 1.000$  implies an integrated process, which is inconsistent with the stationarity assumption of the GARCH framework.

The ACF of squared standardized residuals, shown in Figure 6.2, provides the explanation. All 40 lags fall within the 95% confidence interval, which typically indicates that no ARCH effects remain, but due to statistically insignificant parameters  $\hat{\omega}$ ,  $\hat{\alpha}$ , and  $\hat{\beta}$ , this implies the absence of any statistically significant ARCH effects in the data. When there is no volatility clustering to model, all parameter combinations produce a similar likelihood value, and maximum likelihood estimation cannot identify  $\alpha$  and  $\beta$  separately. This means that the estimates are statistically arbitrary, which is precisely what the wide confidence intervals and insignificant p-values reflect. The ACF of standardized residuals, shown in Figure 6.2, shows a violation at lag 7, suggesting some residual serial dependence in the conditional mean that the zero-mean assumption does not capture. Taken together, these diagnostics indicate that the USDC/USDT pair exhibited no meaningful volatility clustering during the sample period, rendering the benchmark GARCH(1,1) unidentifiable on this data.

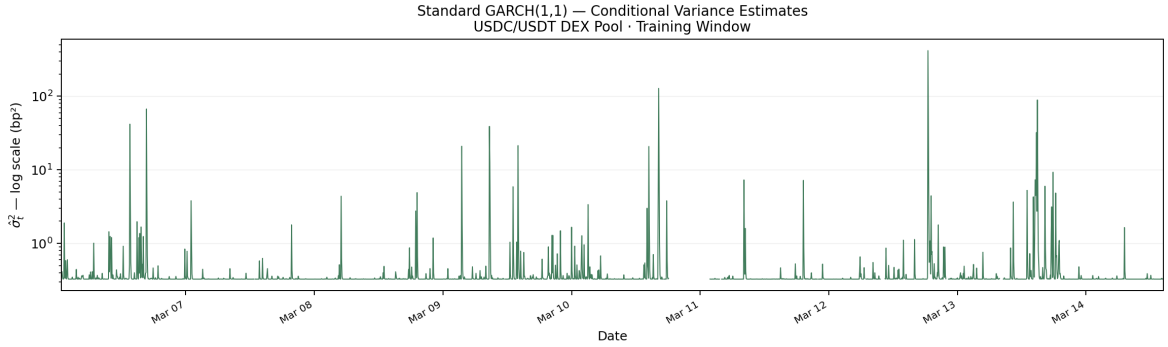
Figure 6.2: ACF of standardized and squared standardized residuals for the benchmark GARCH(1,1).



As a result, the baseline model is extraordinarily sensitive to shocks, overestimates

volatility, and recovers slowly after shocks due to high persistence  $\hat{\alpha} + \hat{\beta} \approx 1.000$ . This is essentially the Integrated GARCH (IGARCH) model proposed by Engle and Bollerslev (1986), implying that shocks contaminate the variance forecast for an extended period, as shown in Figure 6.3, which is inconsistent with the sharp and transient volatility spikes visible in Figure 4.1.

Figure 6.3: Benchmark GARCH(1,1) conditional variance on the training window, showing explosive shock amplification.



### 6.3.2 Hybrid Model Results

The hybrid model is estimated on the same training window using Run 8 from Table 6.3: 3000 epochs, learning rate  $10^{-4}$ , sigmoid output constraint, and weight decay  $10^{-4}$ . The training loss decreases monotonically from  $-0.533$  at epoch 1 to  $-1.408$  at epoch 3000, with the validation loss tracking closely at  $-1.397$ , confirming convergence without overfitting. The best validation NLL of  $-1.397$  is achieved at epoch 3000, indicating that the model had not yet begun to overfit within the training budget.

The time-varying parameters tell a clear story. Across the training window,  $\alpha_t$  has a mean of  $0.000615$  and  $\beta_t$  a mean of  $0.001043$ , with  $\alpha_t + \beta_t \approx 0.0017$  on average. Relative to the bounds of  $a = 0.10$  and  $b = 0.90$ , both parameters are essentially at zero, implying that the model has collapsed to near-constant variance  $\sigma_t^2 \approx \hat{\omega} = 0.2402$  throughout the sample. This is particularly high compared to realized volatility, as shown later in Figure 6.7, because  $\hat{\omega}$  is estimated under the assumption that  $\alpha_t + \beta_t \approx 0.001$ , resulting in the unconditional variance  $\sigma^2 = \frac{\hat{\omega}}{1 - \alpha_t - \beta_t}$  being approximately equal to  $\hat{\omega}$  itself. Had the sum converged closer to 1, the unconditional variance would be substantially lower.

The results are consistent across the validation and test windows, where  $\alpha_t + \beta_t$  averages  $0.005$  and  $0.005$ , respectively, with only occasional transient spikes driven by

individual observations. Both parameters remain essentially constant throughout the sample, and the model reduces to near-constant volatility.

The residual diagnostics mirror those of the benchmark. The ACF of squared standardized residuals, as in Figure 6.4, shows zero violations across all 40 lags, confirming that no ARCH effects remain in the residuals, consistent with there being no ARCH effects to model in the first place. The ACF of standardized residuals, as in Figure 6.5, shows 5 violations at lags 2, 3, 7, 22, and 35, more than the benchmark GARCH, which only shows violations at lag 7, suggesting that the hybrid model captures no additional and arguably less serial dependence in the conditional mean.

Figure 6.4: ACF of squared standardized residuals for the hybrid model, confirming absence of remaining ARCH effects.

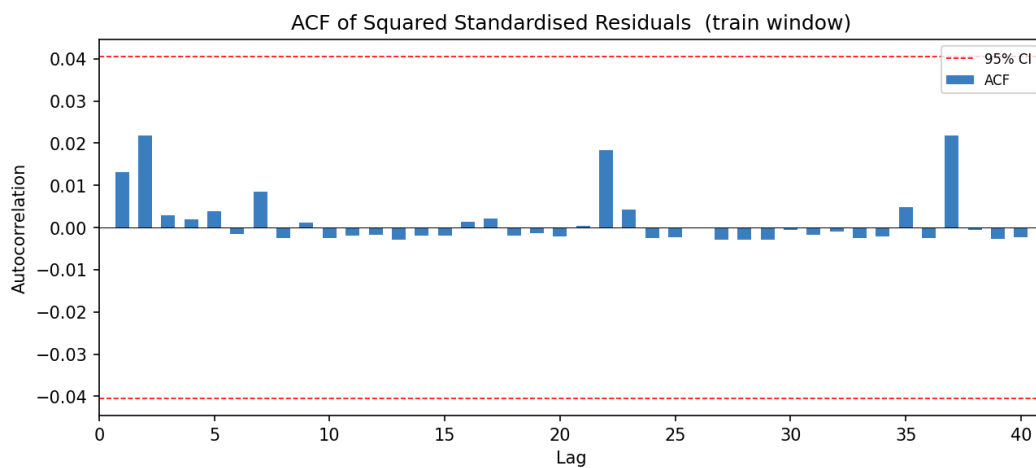
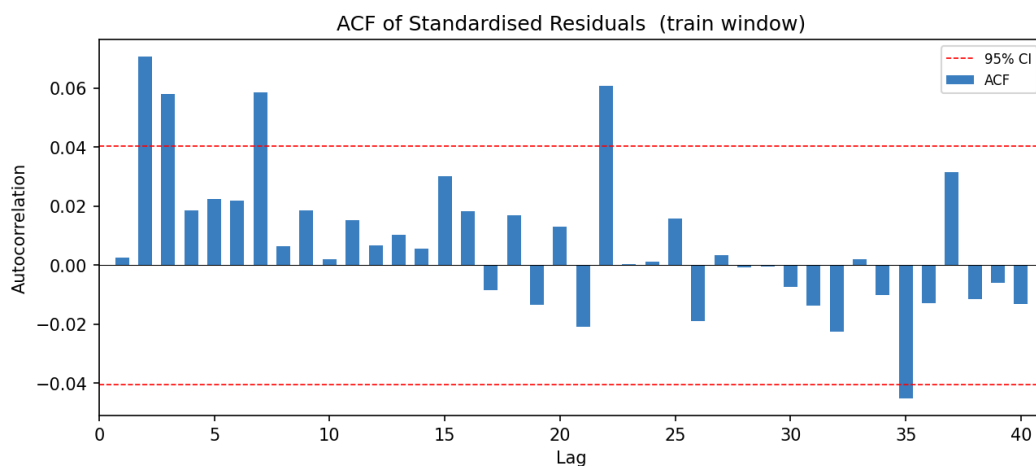


Figure 6.5: ACF of standardized residuals for the hybrid model, showing residual serial dependence.



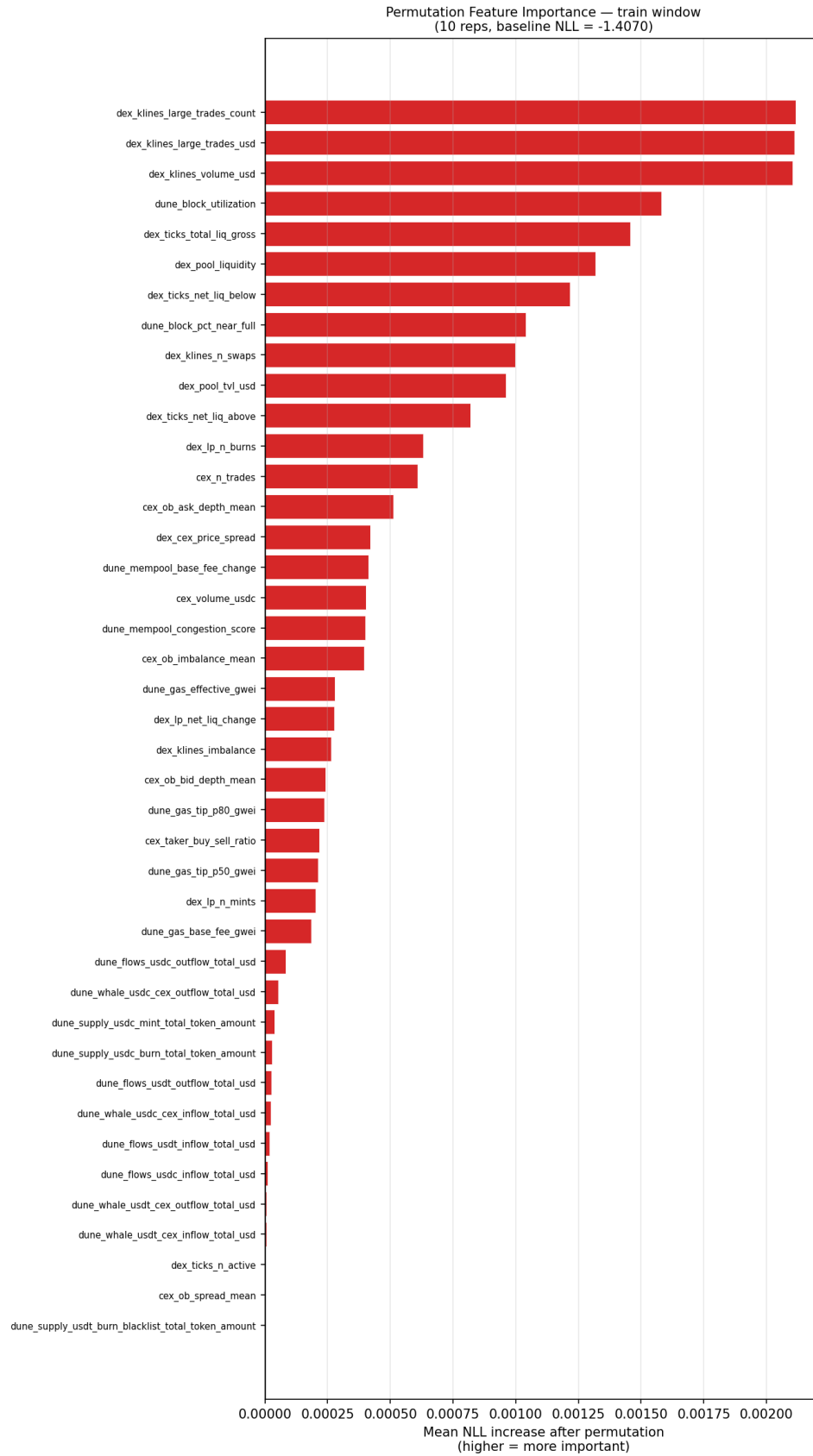
### 6.3.3 Permutation Feature Importance

Despite the parameter collapse, the permutation importance results offer an informative picture of which features the network relies on most when adjusting  $\sigma_t^2$ . Permutation importance is computed by randomly shuffling each feature column across the training window and measuring the resulting increase in the negative log-likelihood relative to the baseline, repeated ten times per feature to reduce noise. A larger NLL increase indicates that the model depends more heavily on that feature.

The results, shown in Figure 6.6, reveal a clear hierarchy. DEX large trade count, large trade volume, and total DEX volume are by far the most important features, each producing an NLL increase roughly three times larger than the next group. This is theoretically coherent: large trades consume a disproportionate share of pool liquidity and generate mechanical price impact, as established in Section 3.2.1, making them the most direct signal of imminent volatility. Block utilization and DEX liquidity variables follow, consistent with the arbitrage friction hypothesis developed in Section 3.3, which states that elevated block utilization raises gas costs and delays cross-market arbitrage, allowing price deviations to persist. CEX trade count ranks thirteenth, which suggests that centralized exchange activity carries some information about DEX volatility beyond what DEX-side variables alone capture, consistent with the price transmission mechanism described in Section 3.3.1.

At the bottom of the ranking, on-chain supply variables (protocol mints, burns, whale transfers, and CEX flows) contribute essentially nothing, with NLL increases near or at zero. This can have two explanations: this either suggests that at the five-minute frequency, macro supply-side events carry no short-term volatility signal and that volatility is driven by execution-side activity rather than fundamental changes in stablecoin supply, or supply changes throughout the data collection window are negligible. Looking at the data, the second hypothesis seems likely, as in a five-minute window only around 10 mint/burn events happen, and both whale transfers and CEX flows are scarce. Two features, `dex_ticks_n_active` and `cex_ob_spread_mean`, register exactly zero importance, indicating the network learned to ignore them entirely.

Figure 6.6: Permutation feature importance for the hybrid model, ranked by mean NLL increase.



### 6.3.4 Volatility Forecasting

Lastly, both models are used in out-of-sample volatility forecasting in order to evaluate whether adding microstructure inputs through the hybrid model adds predictive power. The hybrid model is trained on data between 2026-03-06 00:50:00 and 2026-03-14 14:25:00, and the forecasting origin is 2026-03-14 14:30:00. One-step-ahead forecasts, as described in Section 5.2.1, are used to forecast the next five-minute volatility using the GARCH(1,1) benchmark model, the hybrid sigmoid model from Run 8, and, for comparing the softmax to the sigmoid architecture, the hybrid model from Run 3, as it uses the same hyperparameters (3000 epochs,  $10^{-4}$  learning rate,  $10^{-4}$  weight decay) as Run 8.

As described in Sections 6.3.1 and 6.3.2, the benchmark GARCH(1,1) is destabilized by a high estimate for  $\hat{\alpha} = 0.9267$ , and thus reacts strongly to shocks and does not recover quickly, whereas the hybrid model from Run 8 has collapsed to a near-constant volatility prediction of  $\hat{\sigma}_t^2 \approx \hat{\omega} = 0.2402$ . Thus, both models are unlikely to produce meaningful forecasts given the underlying data, as the GARCH(1,1) will explode and the hybrid model will produce a flat line, as visualized in Figure 6.7.

Nonetheless, the hybrid model from Run 8 performed considerably better than the softmax models from Runs 1 through 5. Run 3, visualized in Figure 6.8, shares the same hyperparameters as Run 8 but produced a substantially worse forecast, illustrating the practical consequence of the softmax collapse described in Section 6.1.2.

Evaluation of all three forecasts against realized five-minute volatility, proxied by  $r_t^2$ , using mean squared error (MSE), root mean squared error (RMSE), and mean absolute error (MAE) shows the benchmark GARCH(1,1) producing slightly better forecasts than the hybrid model from Run 8, both of which substantially outperform the hybrid model from Run 3, as summarized in Table 6.4.

Table 6.4: Out-of-sample forecast accuracy on the test window.

| Model                | MSE   | RMSE  | MAE   |
|----------------------|-------|-------|-------|
| Benchmark GARCH(1,1) | 0.035 | 0.188 | 0.184 |
| Run 8 Hybrid         | 0.058 | 0.240 | 0.240 |
| Run 3 Hybrid         | 772.7 | 27.8  | 3.50  |

Figure 6.7: One-step-ahead variance forecasts for the benchmark GARCH(1,1) and Run 8 hybrid against realized variance.

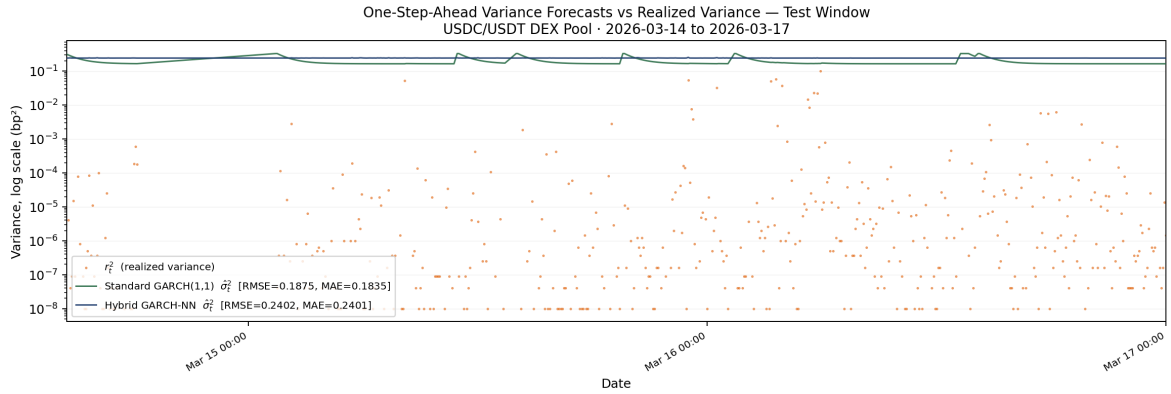
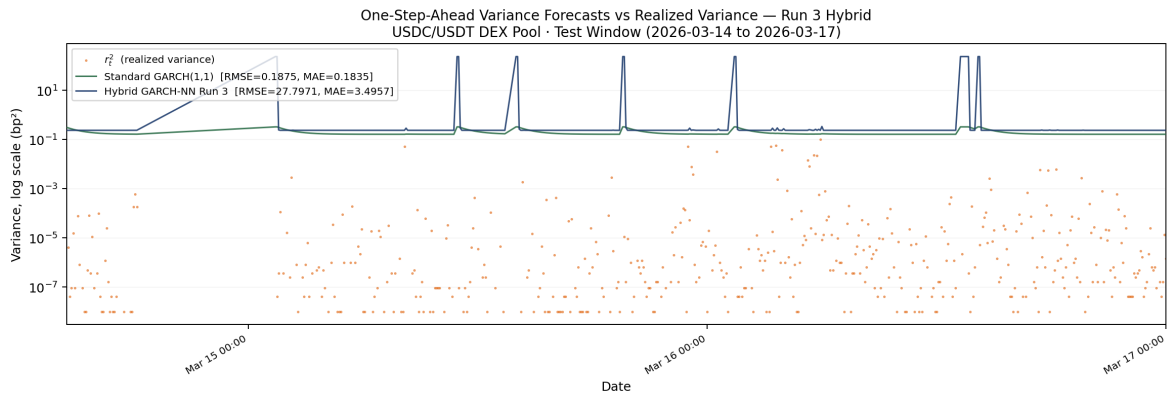


Figure 6.8: One-step-ahead variance forecasts for the benchmark GARCH(1,1) and Run 3 hybrid against realized variance.



## 6.4 Limitations

The results presented in Section 6.3 point to a single underlying cause: the sample period does not contain the regime variation that GARCH-type models require to be identifiable. This section discusses this limitation and its implications for the hybrid architecture more broadly.

### 6.4.1 Regime Coverage and Sample Size

The fundamental limitation of this study is that the 12-day sample period captures the USDC/USDT pair in an exceptionally calm regime. Returns behave near white noise throughout, with no peg stress events, no sustained gas congestion spikes, and no liquidity crises of the kind documented in Section 2.2.3. As the squared residual ACF confirms, there are no statistically significant ARCH effects in the data, meaning that volatility does

not cluster in any detectable way during the sample window. This renders both the benchmark GARCH(1,1) and the hybrid model impossible to use. Without periods of elevated volatility to contrast against calm periods, neither model can learn when parameters should be high versus low.

For the hybrid model specifically, the absence of regime variation means the network receives no training signal that would reward adjusting  $\alpha_t$  or  $\beta_t$  away from a constant baseline. The microstructure features may contain genuine volatility information, which the permutation importance results suggest they do, but that information cannot be extracted from a sample in which volatility itself does not vary. A longer sample spanning multiple market regimes, including episodes of peg deviation such as those documented for USDC in Q3 and Q4 2021 and Tether in Q2 2022, elevated gas congestion, and liquidity disruptions, would provide the contrast necessary for both models to be meaningfully estimated.

#### 6.4.2 Model Complexity and Sample Size

A related limitation concerns the ratio of model complexity to available data. The hybrid model has substantially more parameters than the benchmark GARCH(1,1), introducing a risk of overfitting that is partially mitigated by the absence of regime variation (ironically, the same data problem that prevents the model from learning also prevents it from overfitting). With approximately 2,350 training observations and 41 input features, the effective information per parameter is limited, and the gap structure further reduces the number of clean contiguous sequences available for training. A common rule of thumb suggests a 10:1 ratio of training samples to model parameters, which, given the network’s parameter count, amounts to

| Component                     | Parameters                  |
|-------------------------------|-----------------------------|
| First layer weights + biases  | $41 \times 32 + 32 = 1,344$ |
| Second layer weights + biases | $32 \times 32 + 32 = 1,056$ |
| Output layer weights + biases | $32 \times 2 + 2 = 66$      |
| $\alpha_0$ scalar             | 1                           |
| <b>Total</b>                  | <b>2,467</b>                |

$2,467 \times 10 = 24,670$  observations; around ten times the collected observations. As such, a larger dataset would not only improve regime coverage but also provide the statistical power needed to reliably estimate the network’s parameters.

A natural extension of the current model architecture would be to aggregate the microstructure state inputs to a coarser frequency, such as the previously discussed 30-minute aggregation. This would allow the model to capture changes in microstructure over time, which, especially for liquidity and depth changes, could offer better results. Similarly, an LSTM model could outperform a standard feedforward model, with a similar rationale as the 30-minute aggregation. However, as discussed in Section 6.1.1, frequent gaps in observations due to data collection issues eliminate any possibility of using a model that relies on past state. A meaningful empirical evaluation of hidden state or past aggregation models would therefore require a substantially longer and more continuous collection window than the 12 days available here.

## 7. Conclusion

This thesis set out to examine whether conditioning volatility forecasts on market microstructure variables through a hybrid GARCH-neural network model improves upon a standard GARCH(1,1) benchmark. The empirical answer is that this question cannot be resolved on the available data, but the reasons why are themselves informative results.

The theoretical motivation developed in Chapters 2 and 3 establishes that stablecoin volatility is partly mechanical in origin, with execution frictions, blockchain architecture, and the microstructure of exchanges introducing deviations even in the absence of economic shocks. This motivates a model that uses time-varying GARCH parameters conditioned on microstructure inputs, allowing shock reactivity and volatility persistence to vary with market conditions through a feedforward neural network.

The central finding is that the 12-day USDC/USDT sample exhibits no statistically significant ARCH effects, which renders both models unidentifiable. The benchmark collapses to a near-integrated process, amplifying individual shocks, while the hybrid produces near-constant variance. Out-of-sample forecasting confirms this: both models produce forecasts orders of magnitude away from realized variance. This is a consequence of the sample rather than the model architecture. The pair traded in an exceptionally calm regime throughout, with no peg stress events, liquidity crises, or episodes of sustained congestion, which would be necessary to generate the regime variation that GARCH-type models require.

Despite this, the permutation feature importance results provide theoretically coherent evidence that DEX trading activity, DEX liquidity and block utilization carry the most volatility-relevant information, consistent with the mechanical price impact and arbitrage friction hypotheses developed in Chapters 2 and 3. These findings suggest that the model is identifying those relationships, even if the data does not provide sufficient variation to exploit that learning.

The logical next step is to collect a substantially longer sample spanning multiple months to years and multiple volatility regimes. This sample preferably includes peg stress

events and episodes of larger deviations from parity. Such a sample would provide the necessary contrast between regimes and would require a minimum of ten times the current observations to optimally train the neural network. Minimizing collection gaps would additionally allow for recurrent architectures such as LSTMs to be considered to enable the model to capture changes in liquidity and congestion over time. Lastly, a non-stablecoin asset pair with higher inherent volatility, such as ETH/USDC or WBTC/ETH, could provide greater regime contrast, though at the cost of conflating microstructural volatility with fundamental price uncertainty.

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